Material Characterization Report

NanoVNA UTN Toolkit

2026-10-24 13:12:14

Measurement Details:

- **Sample:** Muestra1_267c
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

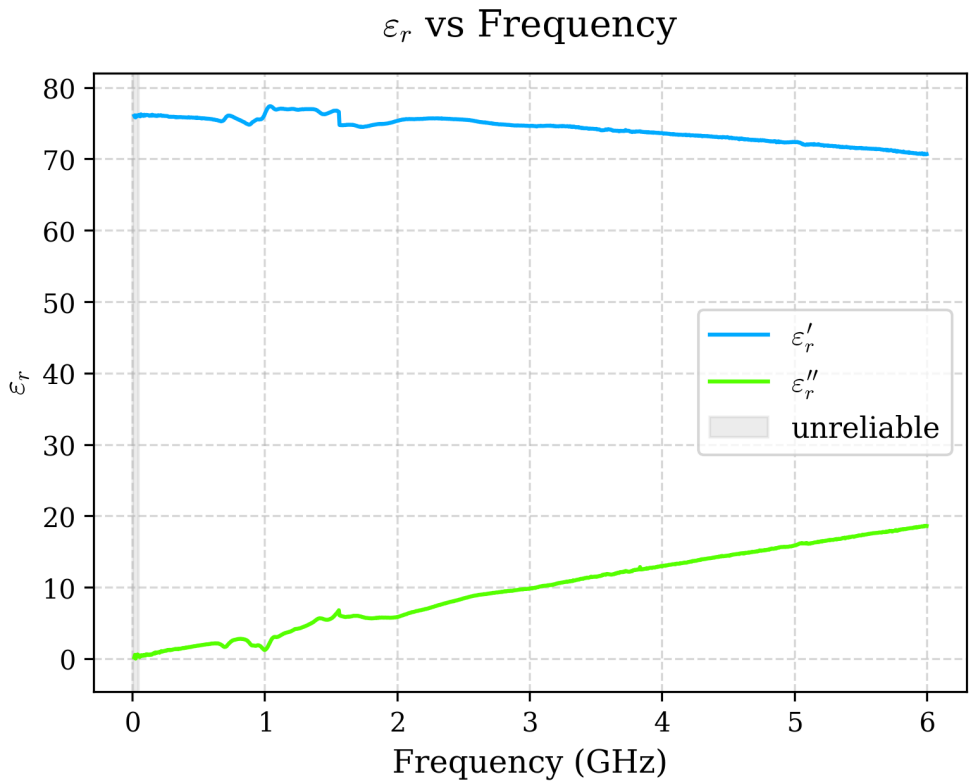
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table).

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	76.0473	0.1670	0.0022
0.020	75.8551	0.5405	0.0071
0.025	75.8176	0.0057	0.0001
0.030	76.0762	0.4658	0.0061
0.035	76.0983	0.4703	0.0062
0.040	76.0120	0.5924	0.0078
0.045	76.0442	0.3940	0.0052
0.050	76.1671	0.4478	0.0059
0.055	76.1076	0.2176	0.0029
0.060	75.9603	0.4497	0.0059
0.065	76.2496	0.4706	0.0062
0.070	76.1738	0.3616	0.0047
0.075	76.0550	0.4841	0.0064
0.080	76.1445	0.4557	0.0060
0.085	76.0242	0.3991	0.0053

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.090	76.1912	0.5314	0.0070
0.095	76.0601	0.4808	0.0063
0.100	76.1062	0.4451	0.0058
0.105	76.1675	0.6110	0.0080
0.110	76.1530	0.5998	0.0079
0.115	76.1507	0.6017	0.0079
0.120	76.1202	0.5155	0.0068
0.125	76.1163	0.5362	0.0070
0.130	76.0389	0.5238	0.0069
0.135	76.0535	0.5980	0.0079
0.140	76.1199	0.5839	0.0077
0.145	76.0469	0.5484	0.0072
0.150	76.1884	0.6441	0.0085
0.155	76.0601	0.6596	0.0087
0.160	76.1002	0.7400	0.0097
0.165	76.1012	0.8007	0.0105
0.170	76.0432	0.7104	0.0093
0.175	76.0967	0.8732	0.0115
0.180	76.0246	0.7817	0.0103
0.185	76.0767	0.8132	0.0107
0.190	76.0279	0.8801	0.0116
0.195	76.1201	0.8825	0.0116
0.200	76.0654	0.9138	0.0120
0.205	76.0792	0.8365	0.0110
0.210	76.0957	0.9757	0.0128
0.215	76.1228	0.9482	0.0125
0.220	76.0910	1.0614	0.0139
0.225	76.0260	0.9832	0.0129
0.230	76.0467	1.0225	0.0134
0.235	76.0626	1.0282	0.0135
0.240	76.0301	1.0777	0.0142
0.245	76.0147	1.1074	0.0146
0.250	76.0178	1.0939	0.0144
0.255	76.0091	1.1048	0.0145
0.260	76.0080	1.1811	0.0155
0.265	76.0005	1.1720	0.0154
0.270	75.9573	1.2242	0.0161
0.275	75.9106	1.2202	0.0161
0.280	75.9045	1.2474	0.0164
0.285	75.9028	1.1833	0.0156
0.290	75.9224	1.2401	0.0163
0.295	75.8958	1.2225	0.0161
0.300	75.9017	1.2346	0.0163
0.305	75.9072	1.2831	0.0169
0.310	75.9131	1.2909	0.0170
0.315	75.9099	1.3117	0.0173

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.320	75.8702	1.3402	0.0177
0.325	75.8499	1.3311	0.0175
0.330	75.8839	1.3751	0.0181
0.335	75.8334	1.3469	0.0178
0.340	75.8319	1.3541	0.0179
0.345	75.8652	1.3232	0.0174
0.350	75.8472	1.3923	0.0184
0.355	75.8497	1.3737	0.0181
0.360	75.8671	1.4056	0.0185
0.365	75.8510	1.3888	0.0183
0.370	75.8550	1.4301	0.0189
0.375	75.8405	1.4483	0.0191
0.380	75.8197	1.4771	0.0195
0.385	75.8333	1.5044	0.0198
0.390	75.8252	1.4673	0.0194
0.395	75.8158	1.5084	0.0199
0.400	75.8075	1.5209	0.0201
0.405	75.8106	1.5682	0.0207
0.410	75.8031	1.5405	0.0203
0.415	75.8114	1.5650	0.0206
0.420	75.7843	1.5742	0.0208
0.425	75.8489	1.6179	0.0213
0.430	75.8276	1.6162	0.0213
0.435	75.7771	1.6145	0.0213
0.440	75.8054	1.6495	0.0218
0.445	75.8188	1.6408	0.0216
0.450	75.7972	1.6942	0.0224
0.455	75.7795	1.6948	0.0224
0.460	75.7925	1.7047	0.0225
0.465	75.7810	1.7506	0.0231
0.470	75.7678	1.7499	0.0231
0.475	75.7783	1.7492	0.0231
0.480	75.7414	1.7755	0.0234
0.485	75.7860	1.7897	0.0236
0.490	75.7318	1.8181	0.0240
0.495	75.7609	1.8352	0.0242
0.500	75.7312	1.8194	0.0240
0.505	75.7380	1.8517	0.0244
0.510	75.7314	1.8619	0.0246
0.515	75.7037	1.8716	0.0247
0.520	75.7112	1.8892	0.0250
0.525	75.6620	1.9178	0.0253
0.530	75.6980	1.9182	0.0253
0.535	75.6940	1.9349	0.0256
0.540	75.6819	1.9586	0.0259
0.545	75.6873	1.9693	0.0260

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.550	75.6532	1.9648	0.0260
0.555	75.6611	2.0045	0.0265
0.560	75.6343	2.0084	0.0266
0.565	75.6471	2.0259	0.0268
0.570	75.6208	2.0129	0.0266
0.575	75.6299	2.0242	0.0268
0.580	75.6197	2.0513	0.0271
0.585	75.6049	2.0554	0.0272
0.590	75.6075	2.0611	0.0273
0.595	75.5862	2.1050	0.0278
0.600	75.5770	2.1243	0.0281
0.605	75.5614	2.1137	0.0280
0.610	75.5191	2.1176	0.0280
0.615	75.5098	2.1314	0.0282
0.620	75.5171	2.1371	0.0283
0.625	75.4886	2.1476	0.0284
0.630	75.4578	2.1280	0.0282
0.635	75.4442	2.1461	0.0284
0.640	75.4406	2.1234	0.0281
0.645	75.4045	2.1391	0.0284
0.650	75.3708	2.1341	0.0283
0.655	75.3395	2.0809	0.0276
0.660	75.3362	2.0483	0.0272
0.665	75.3050	2.0161	0.0268
0.670	75.2635	1.9670	0.0261
0.675	75.2818	1.8980	0.0252
0.680	75.3116	1.8483	0.0245
0.685	75.3421	1.7492	0.0232
0.690	75.4140	1.6873	0.0224
0.695	75.5230	1.6391	0.0217
0.700	75.6461	1.6492	0.0218
0.705	75.7777	1.6637	0.0220
0.710	75.9045	1.7136	0.0226
0.715	75.9754	1.8146	0.0239
0.720	76.0457	1.9349	0.0254
0.725	76.0736	2.0243	0.0266
0.730	76.0650	2.1406	0.0281
0.735	76.0717	2.2140	0.0291
0.740	76.0377	2.3124	0.0304
0.745	76.0204	2.3989	0.0316
0.750	75.9627	2.4489	0.0322
0.755	75.9474	2.5338	0.0334
0.760	75.9088	2.5660	0.0338
0.765	75.8418	2.6035	0.0343
0.770	75.7983	2.6323	0.0347
0.775	75.7554	2.6440	0.0349

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.780	75.7219	2.6702	0.0353
0.785	75.6709	2.7010	0.0357
0.790	75.6303	2.7135	0.0359
0.795	75.5881	2.7347	0.0362
0.800	75.5477	2.7545	0.0365
0.805	75.5043	2.7620	0.0366
0.810	75.4444	2.7621	0.0366
0.815	75.4321	2.7770	0.0368
0.820	75.3572	2.7821	0.0369
0.825	75.3229	2.7612	0.0367
0.830	75.2666	2.7457	0.0365
0.835	75.2337	2.7524	0.0366
0.840	75.1728	2.7284	0.0363
0.845	75.1471	2.7014	0.0359
0.850	75.0880	2.6917	0.0358
0.855	75.0201	2.6493	0.0353
0.860	74.9447	2.6001	0.0347
0.865	74.9210	2.5667	0.0343
0.870	74.8764	2.5093	0.0335
0.875	74.8405	2.3898	0.0319
0.880	74.8022	2.3060	0.0308
0.885	74.8083	2.1887	0.0293
0.890	74.8408	2.0853	0.0279
0.895	74.9112	1.9675	0.0263
0.900	74.9997	1.8832	0.0251
0.905	75.1138	1.8459	0.0246
0.910	75.1718	1.8479	0.0246
0.915	75.2318	1.8454	0.0245
0.920	75.2810	1.8259	0.0243
0.925	75.3447	1.8127	0.0241
0.930	75.4168	1.7995	0.0239
0.935	75.4584	1.8053	0.0239
0.940	75.5293	1.8430	0.0244
0.945	75.5467	1.8717	0.0248
0.950	75.5417	1.8798	0.0249
0.955	75.5241	1.8671	0.0247
0.960	75.5280	1.8245	0.0242
0.965	75.5439	1.7802	0.0236
0.970	75.5477	1.7143	0.0227
0.975	75.5697	1.6107	0.0213
0.980	75.6506	1.5073	0.0199
0.985	75.7571	1.4077	0.0186
0.990	75.9023	1.3376	0.0176
0.995	76.0433	1.2394	0.0163
1.000	76.2701	1.2159	0.0159
1.005	76.4957	1.2545	0.0164

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.010	76.6968	1.3326	0.0174
1.015	76.8789	1.4368	0.0187
1.020	77.0668	1.5694	0.0204
1.025	77.1806	1.7383	0.0225
1.030	77.2976	1.9270	0.0249
1.035	77.3630	2.1087	0.0273
1.040	77.3855	2.3046	0.0298
1.045	77.3751	2.4822	0.0321
1.050	77.3447	2.6501	0.0343
1.055	77.2788	2.7981	0.0362
1.060	77.2172	2.8957	0.0375
1.065	77.1238	2.9747	0.0386
1.070	77.0357	3.0304	0.0393
1.075	76.9751	3.0441	0.0395
1.080	76.9344	3.0470	0.0396
1.085	76.9109	3.0209	0.0393
1.090	76.9146	3.0131	0.0392
1.095	76.9384	3.0424	0.0395
1.100	76.9663	3.0610	0.0398
1.105	77.0016	3.0990	0.0402
1.110	77.0061	3.1539	0.0410
1.115	77.0292	3.2115	0.0417
1.120	77.0465	3.2701	0.0424
1.125	77.0538	3.3238	0.0431
1.130	77.0171	3.3689	0.0437
1.135	77.0450	3.4123	0.0443
1.140	77.0317	3.4745	0.0451
1.145	77.0009	3.5246	0.0458
1.150	76.9848	3.5475	0.0461
1.155	76.9804	3.5996	0.0468
1.160	76.9783	3.6165	0.0470
1.165	76.9551	3.6511	0.0474
1.170	76.9474	3.6740	0.0477
1.175	76.9481	3.7000	0.0481
1.180	76.9543	3.7202	0.0483
1.185	76.9622	3.7619	0.0489
1.190	76.9807	3.7748	0.0490
1.195	76.9772	3.8021	0.0494
1.200	77.0063	3.8501	0.0500
1.205	77.0110	3.9017	0.0507
1.210	77.0085	3.9429	0.0512
1.215	77.0040	4.0013	0.0520
1.220	77.0007	4.0438	0.0525
1.225	76.9665	4.1086	0.0534
1.230	76.9590	4.1536	0.0540
1.235	76.9202	4.1600	0.0541

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.240	76.9058	4.1722	0.0543
1.245	76.9071	4.1802	0.0544
1.250	76.8792	4.1938	0.0546
1.255	76.8830	4.2244	0.0549
1.260	76.8970	4.2366	0.0551
1.265	76.9274	4.2514	0.0553
1.270	76.9265	4.2816	0.0557
1.275	76.9385	4.3194	0.0561
1.280	76.9441	4.3610	0.0567
1.285	76.9730	4.3921	0.0571
1.290	76.9383	4.4414	0.0577
1.295	76.9494	4.4638	0.0580
1.300	76.9716	4.5159	0.0587
1.305	76.9666	4.5400	0.0590
1.310	76.9781	4.5774	0.0595
1.315	76.9739	4.6284	0.0601
1.320	76.9702	4.6531	0.0605
1.325	76.9737	4.7080	0.0612
1.330	76.9783	4.7527	0.0617
1.335	76.9771	4.8105	0.0625
1.340	76.9721	4.8404	0.0629
1.345	76.9755	4.8867	0.0635
1.350	76.9825	4.9608	0.0644
1.355	76.9695	4.9768	0.0647
1.360	76.9702	5.0455	0.0656
1.365	76.9745	5.0956	0.0662
1.370	76.9588	5.1584	0.0670
1.375	76.9436	5.2333	0.0680
1.380	76.9204	5.3026	0.0689
1.385	76.8963	5.3878	0.0701
1.390	76.8497	5.4509	0.0709
1.395	76.8039	5.4964	0.0716
1.400	76.7430	5.5715	0.0726
1.405	76.6629	5.6139	0.0732
1.410	76.6020	5.6395	0.0736
1.415	76.5292	5.6642	0.0740
1.420	76.4619	5.6495	0.0739
1.425	76.3843	5.6371	0.0738
1.430	76.3381	5.6319	0.0738
1.435	76.2789	5.6132	0.0736
1.440	76.2727	5.5685	0.0730
1.445	76.2752	5.5342	0.0726
1.450	76.2643	5.5091	0.0722
1.455	76.2510	5.4813	0.0719
1.460	76.2910	5.4586	0.0716
1.465	76.3117	5.4463	0.0714

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.470	76.3461	5.4503	0.0714
1.475	76.3832	5.4433	0.0713
1.480	76.4173	5.4598	0.0714
1.485	76.4607	5.4845	0.0717
1.490	76.5022	5.5203	0.0722
1.495	76.5301	5.5564	0.0726
1.500	76.5735	5.6004	0.0731
1.505	76.6230	5.6599	0.0739
1.510	76.6517	5.7136	0.0745
1.515	76.6978	5.8035	0.0757
1.520	76.7402	5.8686	0.0765
1.525	76.7462	5.9762	0.0779
1.530	76.7446	6.0694	0.0791
1.535	76.7567	6.1958	0.0807
1.540	76.7608	6.2808	0.0818
1.545	76.7564	6.3922	0.0833
1.550	76.7177	6.5278	0.0851
1.555	76.6532	6.6226	0.0864
1.560	76.5973	6.7831	0.0886
1.565	74.7637	6.0675	0.0812
1.570	74.7256	6.0261	0.0806
1.575	74.7293	5.9805	0.0800
1.580	74.7367	5.9614	0.0798
1.585	74.7111	5.9451	0.0796
1.590	74.7305	5.9141	0.0791
1.595	74.7201	5.8824	0.0787
1.600	74.7405	5.8856	0.0787
1.605	74.7450	5.8459	0.0782
1.610	74.7359	5.8336	0.0781
1.615	74.7622	5.8418	0.0781
1.620	74.7866	5.8691	0.0785
1.625	74.7752	5.8793	0.0786
1.630	74.7871	5.8666	0.0784
1.635	74.7874	5.8974	0.0789
1.640	74.7879	5.8745	0.0785
1.645	74.7927	5.8922	0.0788
1.650	74.7812	5.9045	0.0790
1.655	74.7782	5.9372	0.0794
1.660	74.7728	5.9430	0.0795
1.665	74.7862	5.9756	0.0799
1.670	74.7483	6.0044	0.0803
1.675	74.7285	6.0003	0.0803
1.680	74.7159	6.0011	0.0803
1.685	74.6748	5.9982	0.0803
1.690	74.6585	6.0053	0.0804
1.695	74.6366	5.9995	0.0804

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.700	74.5886	6.0016	0.0805
1.705	74.5706	6.0025	0.0805
1.710	74.5459	5.9689	0.0801
1.715	74.5157	5.9426	0.0797
1.720	74.5148	5.9335	0.0796
1.725	74.4762	5.9161	0.0794
1.730	74.5190	5.8931	0.0791
1.735	74.4788	5.8306	0.0783
1.740	74.4784	5.8247	0.0782
1.745	74.4845	5.8054	0.0779
1.750	74.5086	5.7645	0.0774
1.755	74.5301	5.7614	0.0773
1.760	74.5447	5.7182	0.0767
1.765	74.5573	5.7060	0.0765
1.770	74.5484	5.7089	0.0766
1.775	74.5815	5.6967	0.0764
1.780	74.6028	5.6760	0.0761
1.785	74.6115	5.6747	0.0761
1.790	74.6239	5.6647	0.0759
1.795	74.6692	5.6605	0.0758
1.800	74.6848	5.6574	0.0757
1.805	74.7173	5.6420	0.0755
1.810	74.7245	5.6535	0.0757
1.815	74.7534	5.6509	0.0756
1.820	74.7682	5.6609	0.0757
1.825	74.7926	5.6663	0.0758
1.830	74.8032	5.6837	0.0760
1.835	74.8085	5.6882	0.0760
1.840	74.8353	5.6838	0.0760
1.845	74.8489	5.7329	0.0766
1.850	74.8572	5.7068	0.0762
1.855	74.8765	5.7267	0.0765
1.860	74.8922	5.7087	0.0762
1.865	74.8808	5.7677	0.0770
1.870	74.9116	5.7492	0.0767
1.875	74.8934	5.7400	0.0766
1.880	74.9139	5.7507	0.0768
1.885	74.8866	5.7684	0.0770
1.890	74.9205	5.7595	0.0769
1.895	74.9012	5.7414	0.0767
1.900	74.9118	5.7608	0.0769
1.905	74.9288	5.7454	0.0767
1.910	74.9330	5.7561	0.0768
1.915	74.9385	5.7598	0.0769
1.920	74.9633	5.7415	0.0766
1.925	74.9643	5.7376	0.0765

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.930	74.9788	5.7538	0.0767
1.935	75.0114	5.7439	0.0766
1.940	75.0318	5.7272	0.0763
1.945	75.0453	5.7287	0.0763
1.950	75.0567	5.7299	0.0763
1.955	75.0972	5.7384	0.0764
1.960	75.1039	5.7417	0.0765
1.965	75.1266	5.7515	0.0766
1.970	75.1591	5.7446	0.0764
1.975	75.2033	5.7682	0.0767
1.980	75.2123	5.7782	0.0768
1.985	75.2541	5.7699	0.0767
1.990	75.2755	5.7969	0.0770
1.995	75.2983	5.8134	0.0772
2.000	75.3218	5.8267	0.0774
2.005	75.3573	5.8636	0.0778
2.010	75.3580	5.8725	0.0779
2.015	75.3923	5.9071	0.0784
2.020	75.4318	5.9225	0.0785
2.025	75.4421	5.9659	0.0791
2.030	75.4564	5.9965	0.0795
2.035	75.4948	6.0205	0.0797
2.040	75.5075	6.0589	0.0802
2.045	75.5021	6.0726	0.0804
2.050	75.5393	6.1361	0.0812
2.055	75.5294	6.1612	0.0816
2.060	75.5451	6.1811	0.0818
2.065	75.5745	6.2127	0.0822
2.070	75.5845	6.2458	0.0826
2.075	75.5687	6.2639	0.0829
2.080	75.5672	6.3189	0.0836
2.085	75.5817	6.3583	0.0841
2.090	75.5874	6.3828	0.0844
2.095	75.5958	6.4128	0.0848
2.100	75.5735	6.4393	0.0852
2.105	75.5809	6.4536	0.0854
2.110	75.5816	6.5081	0.0861
2.115	75.5807	6.5168	0.0862
2.120	75.5863	6.5287	0.0864
2.125	75.5700	6.5520	0.0867
2.130	75.5606	6.5707	0.0870
2.135	75.5786	6.5971	0.0873
2.140	75.5889	6.6404	0.0878
2.145	75.5865	6.6639	0.0882
2.150	75.5940	6.6837	0.0884
2.155	75.6020	6.6791	0.0883

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.160	75.5974	6.7118	0.0888
2.165	75.5917	6.7432	0.0892
2.170	75.5931	6.7636	0.0895
2.175	75.6088	6.7715	0.0896
2.180	75.6098	6.7877	0.0898
2.185	75.6046	6.8093	0.0901
2.190	75.6210	6.8469	0.0905
2.195	75.6044	6.8631	0.0908
2.200	75.6381	6.8846	0.0910
2.205	75.6357	6.8957	0.0912
2.210	75.6395	6.9428	0.0918
2.215	75.6374	6.9619	0.0920
2.220	75.6546	6.9590	0.0920
2.225	75.6599	6.9938	0.0924
2.230	75.6552	7.0073	0.0926
2.235	75.6801	7.0297	0.0929
2.240	75.6729	7.0820	0.0936
2.245	75.6782	7.0996	0.0938
2.250	75.6817	7.1368	0.0943
2.255	75.6741	7.1481	0.0945
2.260	75.6825	7.1848	0.0949
2.265	75.6682	7.2254	0.0955
2.270	75.6881	7.2391	0.0956
2.275	75.6794	7.2587	0.0959
2.280	75.6746	7.2960	0.0964
2.285	75.6658	7.3507	0.0971
2.290	75.6816	7.3556	0.0972
2.295	75.6812	7.3874	0.0976
2.300	75.6798	7.4128	0.0979
2.305	75.6869	7.4486	0.0984
2.310	75.6617	7.4851	0.0989
2.315	75.6834	7.5089	0.0992
2.320	75.6621	7.5295	0.0995
2.325	75.6676	7.5260	0.0995
2.330	75.6727	7.5766	0.1001
2.335	75.6373	7.6083	0.1006
2.340	75.6732	7.6281	0.1008
2.345	75.6474	7.6487	0.1011
2.350	75.6461	7.6965	0.1017
2.355	75.6503	7.7191	0.1020
2.360	75.6363	7.7314	0.1022
2.365	75.6366	7.7763	0.1028
2.370	75.6343	7.7686	0.1027
2.375	75.6193	7.8119	0.1033
2.380	75.6303	7.8400	0.1037
2.385	75.6119	7.8696	0.1041

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.390	75.6034	7.8927	0.1044
2.395	75.6221	7.9190	0.1047
2.400	75.6131	7.9423	0.1050
2.405	75.5891	7.9737	0.1055
2.410	75.5910	7.9798	0.1056
2.415	75.5874	8.0106	0.1060
2.420	75.5766	8.0172	0.1061
2.425	75.5719	8.0639	0.1067
2.430	75.5558	8.0447	0.1065
2.435	75.5710	8.0996	0.1072
2.440	75.5713	8.1273	0.1075
2.445	75.5579	8.1388	0.1077
2.450	75.5653	8.1683	0.1081
2.455	75.5445	8.1809	0.1083
2.460	75.5487	8.2165	0.1088
2.465	75.5483	8.2446	0.1091
2.470	75.5255	8.2756	0.1096
2.475	75.5359	8.2742	0.1095
2.480	75.5248	8.3160	0.1101
2.485	75.5198	8.3434	0.1105
2.490	75.5174	8.3593	0.1107
2.495	75.5226	8.4079	0.1113
2.500	75.4989	8.4316	0.1117
2.505	75.4991	8.4528	0.1120
2.510	75.4894	8.4727	0.1122
2.515	75.4915	8.5070	0.1127
2.520	75.4722	8.5276	0.1130
2.525	75.4701	8.5549	0.1134
2.530	75.4699	8.5583	0.1134
2.535	75.4518	8.5964	0.1139
2.540	75.4421	8.6333	0.1144
2.545	75.4486	8.6480	0.1146
2.550	75.4294	8.6703	0.1149
2.555	75.4071	8.7031	0.1154
2.560	75.3895	8.7236	0.1157
2.565	75.3863	8.7508	0.1161
2.570	75.3686	8.7605	0.1162
2.575	75.3498	8.7955	0.1167
2.580	75.3458	8.8067	0.1169
2.585	75.3096	8.8316	0.1173
2.590	75.2976	8.8423	0.1174
2.595	75.2900	8.8488	0.1175
2.600	75.2760	8.8743	0.1179
2.605	75.2337	8.8749	0.1180
2.610	75.2341	8.9090	0.1184
2.615	75.2124	8.9345	0.1188

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.620	75.1917	8.9472	0.1190
2.625	75.1740	8.9497	0.1191
2.630	75.1699	8.9687	0.1193
2.635	75.1610	8.9629	0.1192
2.640	75.1411	8.9931	0.1197
2.645	75.1424	8.9927	0.1197
2.650	75.1162	9.0056	0.1199
2.655	75.1072	9.0429	0.1204
2.660	75.1095	9.0210	0.1201
2.665	75.1214	9.0464	0.1204
2.670	75.0680	9.0443	0.1205
2.675	75.0868	9.0547	0.1206
2.680	75.0687	9.0919	0.1211
2.685	75.0508	9.0922	0.1211
2.690	75.0344	9.1161	0.1215
2.695	75.0217	9.1214	0.1216
2.700	75.0102	9.1379	0.1218
2.705	74.9849	9.1561	0.1221
2.710	75.0105	9.1639	0.1222
2.715	74.9763	9.1695	0.1223
2.720	74.9788	9.1952	0.1226
2.725	74.9252	9.2250	0.1231
2.730	74.9344	9.2148	0.1230
2.735	74.9210	9.2227	0.1231
2.740	74.9051	9.2428	0.1234
2.745	74.8930	9.2621	0.1237
2.750	74.8894	9.2777	0.1239
2.755	74.8981	9.2747	0.1238
2.760	74.8766	9.2721	0.1238
2.765	74.8470	9.2992	0.1242
2.770	74.8386	9.3199	0.1245
2.775	74.8401	9.3139	0.1245
2.780	74.8352	9.3373	0.1248
2.785	74.8498	9.3515	0.1249
2.790	74.8159	9.3361	0.1248
2.795	74.8315	9.3792	0.1253
2.800	74.8328	9.3740	0.1253
2.805	74.8147	9.4005	0.1257
2.810	74.8110	9.4061	0.1257
2.815	74.8141	9.4534	0.1264
2.820	74.8005	9.4392	0.1262
2.825	74.7825	9.4208	0.1260
2.830	74.7752	9.4598	0.1265
2.835	74.7734	9.4691	0.1266
2.840	74.7696	9.4860	0.1269
2.845	74.7697	9.4728	0.1267

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.850	74.7518	9.4869	0.1269
2.855	74.7402	9.5110	0.1273
2.860	74.7396	9.5004	0.1271
2.865	74.7118	9.5064	0.1272
2.870	74.7162	9.5051	0.1272
2.875	74.7098	9.5297	0.1276
2.880	74.7036	9.5454	0.1278
2.885	74.6875	9.5690	0.1281
2.890	74.6758	9.5762	0.1282
2.895	74.6909	9.6004	0.1285
2.900	74.6844	9.6020	0.1286
2.905	74.6662	9.6161	0.1288
2.910	74.6957	9.6339	0.1290
2.915	74.6932	9.6469	0.1292
2.920	74.6777	9.6722	0.1295
2.925	74.6728	9.6670	0.1295
2.930	74.6731	9.7027	0.1299
2.935	74.6793	9.6943	0.1298
2.940	74.6803	9.7150	0.1301
2.945	74.6631	9.7302	0.1303
2.950	74.6693	9.6972	0.1299
2.955	74.6735	9.7221	0.1302
2.960	74.6650	9.7569	0.1307
2.965	74.6515	9.7476	0.1306
2.970	74.6511	9.7573	0.1307
2.975	74.6394	9.7534	0.1307
2.980	74.6565	9.7662	0.1308
2.985	74.6354	9.7918	0.1312
2.990	74.6354	9.7918	0.1312
2.995	74.6196	9.7822	0.1311
3.000	74.6198	9.8268	0.1317
3.005	74.6177	9.8482	0.1320
3.010	74.6521	9.8456	0.1319
3.015	74.6381	9.8418	0.1319
3.020	74.6208	9.8963	0.1326
3.025	74.6452	9.8763	0.1323
3.030	74.6151	9.8873	0.1325
3.035	74.6039	9.9075	0.1328
3.040	74.5972	9.9408	0.1333
3.045	74.6079	9.9110	0.1328
3.050	74.5688	9.9616	0.1336
3.055	74.5475	9.9908	0.1340
3.060	74.5939	10.0128	0.1342
3.065	74.6142	10.0074	0.1341
3.070	74.6151	10.0538	0.1347
3.075	74.6148	10.0966	0.1353

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.080	74.6186	10.1066	0.1354
3.085	74.6495	10.1286	0.1357
3.090	74.6248	10.1414	0.1359
3.095	74.6142	10.1436	0.1359
3.100	74.6154	10.1963	0.1367
3.105	74.6258	10.1907	0.1366
3.110	74.6229	10.2124	0.1369
3.115	74.6348	10.2212	0.1369
3.120	74.6337	10.2145	0.1369
3.125	74.6582	10.2322	0.1371
3.130	74.6442	10.2290	0.1370
3.135	74.6583	10.2582	0.1374
3.140	74.6537	10.2943	0.1379
3.145	74.6504	10.2510	0.1373
3.150	74.6660	10.2960	0.1379
3.155	74.6355	10.3255	0.1383
3.160	74.6715	10.3059	0.1380
3.165	74.6329	10.3231	0.1383
3.170	74.6241	10.3586	0.1388
3.175	74.6180	10.3293	0.1384
3.180	74.6534	10.4035	0.1394
3.185	74.5953	10.4089	0.1395
3.190	74.5819	10.4220	0.1397
3.195	74.6014	10.4606	0.1402
3.200	74.5571	10.4954	0.1408
3.205	74.5755	10.4985	0.1408
3.210	74.5694	10.5486	0.1415
3.215	74.5790	10.5515	0.1415
3.220	74.5535	10.5943	0.1421
3.225	74.5851	10.5946	0.1420
3.230	74.6014	10.5937	0.1420
3.235	74.5923	10.6340	0.1426
3.240	74.5812	10.6517	0.1428
3.245	74.6170	10.6623	0.1429
3.250	74.5755	10.6590	0.1429
3.255	74.5994	10.6982	0.1434
3.260	74.6133	10.7105	0.1435
3.265	74.5804	10.7269	0.1438
3.270	74.5692	10.7435	0.1441
3.275	74.5548	10.7673	0.1444
3.280	74.5778	10.7607	0.1443
3.285	74.5938	10.8100	0.1449
3.290	74.5802	10.7748	0.1445
3.295	74.5203	10.8004	0.1449
3.300	74.5552	10.8418	0.1454
3.305	74.5917	10.8206	0.1451

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.310	74.5083	10.8674	0.1459
3.315	74.5037	10.8589	0.1457
3.320	74.5295	10.8779	0.1460
3.325	74.4667	10.8625	0.1459
3.330	74.5068	10.9783	0.1473
3.335	74.5359	10.9557	0.1470
3.340	74.5042	10.9527	0.1470
3.345	74.4604	10.9649	0.1473
3.350	74.4202	11.0086	0.1479
3.355	74.4560	11.0238	0.1481
3.360	74.4482	11.0651	0.1486
3.365	74.4524	11.0547	0.1485
3.370	74.4071	11.0841	0.1490
3.375	74.4366	11.0774	0.1488
3.380	74.3731	11.1339	0.1497
3.385	74.3737	11.1653	0.1501
3.390	74.3366	11.1939	0.1506
3.395	74.3642	11.1827	0.1504
3.400	74.3177	11.1949	0.1506
3.405	74.3297	11.1846	0.1505
3.410	74.2854	11.2466	0.1514
3.415	74.2985	11.2636	0.1516
3.420	74.3031	11.3068	0.1522
3.425	74.2878	11.3009	0.1521
3.430	74.3020	11.3098	0.1522
3.435	74.2964	11.3579	0.1529
3.440	74.2579	11.3298	0.1526
3.445	74.2361	11.3282	0.1526
3.450	74.2160	11.3500	0.1529
3.455	74.2576	11.3923	0.1534
3.460	74.2507	11.4125	0.1537
3.465	74.2287	11.4861	0.1547
3.470	74.2359	11.4191	0.1538
3.475	74.2358	11.4358	0.1540
3.480	74.2233	11.4458	0.1542
3.485	74.2156	11.4726	0.1546
3.490	74.2057	11.4673	0.1545
3.495	74.1963	11.4989	0.1550
3.500	74.1965	11.4758	0.1547
3.505	74.2153	11.4613	0.1544
3.510	74.1452	11.5032	0.1551
3.515	74.1341	11.5007	0.1551
3.520	74.1287	11.5478	0.1558
3.525	74.1447	11.5605	0.1559
3.530	74.0874	11.5891	0.1564
3.535	74.0622	11.5791	0.1563

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.540	74.0099	11.6280	0.1571
3.545	74.0226	11.6228	0.1570
3.550	74.0023	11.7327	0.1585
3.555	74.0225	11.6928	0.1580
3.560	74.0187	11.7546	0.1588
3.565	74.0343	11.7877	0.1592
3.570	74.0638	11.8034	0.1594
3.575	74.1303	11.7428	0.1584
3.580	74.1310	11.8331	0.1596
3.585	74.0701	11.8689	0.1602
3.590	74.1375	11.8590	0.1600
3.595	74.1739	11.8244	0.1594
3.600	74.1420	11.8325	0.1596
3.605	74.1101	11.8018	0.1592
3.610	74.1206	11.7967	0.1592
3.615	74.1009	11.7707	0.1588
3.620	74.0354	11.7735	0.1590
3.625	74.0278	11.7983	0.1594
3.630	73.9602	11.8222	0.1598
3.635	73.9620	11.8226	0.1598
3.640	73.9550	11.8392	0.1601
3.645	73.9579	11.9178	0.1611
3.650	73.9584	11.9309	0.1613
3.655	73.9433	11.9135	0.1611
3.660	73.9339	11.9934	0.1622
3.665	73.9512	12.0276	0.1626
3.670	73.9361	12.0423	0.1629
3.675	73.8750	11.9999	0.1624
3.680	73.8771	12.0842	0.1636
3.685	73.9312	12.1316	0.1641
3.690	73.9430	12.1059	0.1637
3.695	73.9189	12.1258	0.1640
3.700	73.9188	12.1490	0.1644
3.705	73.9216	12.1151	0.1639
3.710	73.9028	12.1918	0.1650
3.715	73.9294	12.2054	0.1651
3.720	73.9171	12.1762	0.1647
3.725	74.0495	12.2810	0.1658
3.730	73.9523	12.2304	0.1654
3.735	73.9399	12.2452	0.1656
3.740	73.9568	12.1937	0.1649
3.745	73.9437	12.1902	0.1649
3.750	73.9455	12.1742	0.1646
3.755	73.9059	12.1767	0.1648
3.760	73.8347	12.1858	0.1650
3.765	73.8346	12.2442	0.1658

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.770	73.8032	12.2758	0.1663
3.775	73.7582	12.2458	0.1660
3.780	73.8057	12.3582	0.1674
3.785	73.8302	12.3948	0.1679
3.790	73.8124	12.3923	0.1679
3.795	73.7917	12.4468	0.1687
3.800	73.8069	12.4279	0.1684
3.805	73.8604	12.4430	0.1685
3.810	73.8408	12.4728	0.1689
3.815	73.8391	12.4617	0.1688
3.820	73.8428	12.4929	0.1692
3.825	73.8567	12.4984	0.1692
3.830	73.8612	12.5186	0.1695
3.835	73.8730	12.8337	0.1737
3.840	73.8050	12.5138	0.1696
3.845	73.8038	12.4920	0.1693
3.850	73.8042	12.5228	0.1697
3.855	73.7812	12.5180	0.1697
3.860	73.7724	12.5526	0.1702
3.865	73.7765	12.5576	0.1702
3.870	73.7482	12.5883	0.1707
3.875	73.7545	12.5915	0.1707
3.880	73.7689	12.6124	0.1710
3.885	73.7192	12.6754	0.1719
3.890	73.7303	12.6136	0.1711
3.895	73.7617	12.6143	0.1710
3.900	73.7099	12.6579	0.1717
3.905	73.7317	12.7118	0.1724
3.910	73.7063	12.7224	0.1726
3.915	73.7018	12.7327	0.1728
3.920	73.7210	12.7202	0.1725
3.925	73.6906	12.7071	0.1724
3.930	73.7087	12.7524	0.1730
3.935	73.6554	12.7762	0.1735
3.940	73.7086	12.7706	0.1733
3.945	73.6798	12.7647	0.1732
3.950	73.6776	12.7824	0.1735
3.955	73.6595	12.8320	0.1742
3.960	73.6636	12.8512	0.1745
3.965	73.6417	12.8359	0.1743
3.970	73.6496	12.8972	0.1751
3.975	73.6652	12.8976	0.1751
3.980	73.6250	12.9081	0.1753
3.985	73.6160	12.9175	0.1755
3.990	73.6229	12.9662	0.1761
3.995	73.6237	12.9439	0.1758

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.000	73.6239	12.9787	0.1763
4.005	73.5913	12.9565	0.1761
4.010	73.5382	12.9911	0.1767
4.015	73.5872	13.0283	0.1770
4.020	73.5638	13.0320	0.1772
4.025	73.5553	13.0033	0.1768
4.030	73.5710	13.0709	0.1777
4.035	73.5382	13.0733	0.1778
4.040	73.5450	13.0735	0.1778
4.045	73.5263	13.0932	0.1781
4.050	73.5217	13.1349	0.1787
4.055	73.5199	13.1406	0.1787
4.060	73.5063	13.1452	0.1788
4.065	73.4556	13.1580	0.1791
4.070	73.4675	13.1670	0.1792
4.075	73.4679	13.1922	0.1796
4.080	73.4766	13.2268	0.1800
4.085	73.4215	13.2224	0.1801
4.090	73.4617	13.2306	0.1801
4.095	73.4638	13.2344	0.1801
4.100	73.4236	13.2708	0.1807
4.105	73.4085	13.2346	0.1803
4.110	73.4185	13.3049	0.1812
4.115	73.4045	13.3168	0.1814
4.120	73.4173	13.2998	0.1812
4.125	73.3706	13.2969	0.1812
4.130	73.4055	13.3174	0.1814
4.135	73.3904	13.3072	0.1813
4.140	73.3887	13.3784	0.1823
4.145	73.3879	13.3751	0.1823
4.150	73.3819	13.3690	0.1822
4.155	73.3577	13.3959	0.1826
4.160	73.3507	13.4087	0.1828
4.165	73.3774	13.4376	0.1831
4.170	73.3743	13.4627	0.1835
4.175	73.3721	13.4345	0.1831
4.180	73.3329	13.4588	0.1835
4.185	73.3729	13.4508	0.1833
4.190	73.3117	13.4652	0.1837
4.195	73.3435	13.5149	0.1843
4.200	73.3433	13.5380	0.1846
4.205	73.3646	13.5344	0.1845
4.210	73.3518	13.5832	0.1852
4.215	73.2771	13.5884	0.1854
4.220	73.3010	13.6046	0.1856
4.225	73.3068	13.5852	0.1853

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.230	73.3040	13.6604	0.1864
4.235	73.3105	13.6868	0.1867
4.240	73.2891	13.6849	0.1867
4.245	73.2795	13.6750	0.1866
4.250	73.2896	13.7159	0.1871
4.255	73.2541	13.7186	0.1873
4.260	73.2786	13.7158	0.1872
4.265	73.2827	13.7462	0.1876
4.270	73.2486	13.7895	0.1883
4.275	73.2696	13.7956	0.1883
4.280	73.2324	13.8310	0.1889
4.285	73.2570	13.8153	0.1886
4.290	73.2271	13.8589	0.1893
4.295	73.2668	13.8431	0.1889
4.300	73.2246	13.9205	0.1901
4.305	73.1713	13.9295	0.1904
4.310	73.2019	13.9194	0.1902
4.315	73.2076	13.9453	0.1905
4.320	73.2212	13.9419	0.1904
4.325	73.2013	13.9774	0.1909
4.330	73.2063	13.9732	0.1909
4.335	73.2037	13.9937	0.1912
4.340	73.1882	14.0226	0.1916
4.345	73.2080	14.0402	0.1918
4.350	73.1711	14.0468	0.1920
4.355	73.1699	14.0483	0.1920
4.360	73.1647	14.0681	0.1923
4.365	73.1507	14.0643	0.1923
4.370	73.1291	14.0818	0.1926
4.375	73.1607	14.0917	0.1926
4.380	73.1197	14.1171	0.1931
4.385	73.1047	14.1529	0.1936
4.390	73.1089	14.1524	0.1936
4.395	73.1029	14.1459	0.1935
4.400	73.0849	14.2019	0.1943
4.405	73.0800	14.2030	0.1943
4.410	73.1048	14.2279	0.1946
4.415	73.0901	14.2415	0.1948
4.420	73.0687	14.2426	0.1949
4.425	73.0242	14.2415	0.1950
4.430	73.0364	14.3064	0.1959
4.435	73.0583	14.2934	0.1956
4.440	73.0508	14.3164	0.1960
4.445	73.0540	14.3646	0.1966
4.450	73.0194	14.3139	0.1960
4.455	72.9985	14.3959	0.1972

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.460	73.0784	14.3458	0.1963
4.465	73.0496	14.3359	0.1962
4.470	73.0486	14.3765	0.1968
4.475	73.0441	14.3758	0.1968
4.480	73.0073	14.3994	0.1972
4.485	73.0085	14.3800	0.1970
4.490	73.0411	14.4133	0.1973
4.495	73.0587	14.4333	0.1976
4.500	73.0153	14.4215	0.1975
4.505	72.9988	14.4575	0.1981
4.510	72.9217	14.4098	0.1976
4.515	72.9708	14.4850	0.1985
4.520	72.9611	14.5272	0.1991
4.525	72.9884	14.5019	0.1987
4.530	72.8919	14.5688	0.1999
4.535	72.9051	14.5723	0.1999
4.540	72.9031	14.6024	0.2003
4.545	72.9057	14.5876	0.2001
4.550	72.8843	14.6132	0.2005
4.555	72.8675	14.6124	0.2005
4.560	72.8820	14.6483	0.2010
4.565	72.8933	14.6321	0.2007
4.570	72.8628	14.6636	0.2012
4.575	72.8437	14.6731	0.2014
4.580	72.8503	14.6568	0.2012
4.585	72.7418	14.6565	0.2015
4.590	72.8255	14.7339	0.2023
4.595	72.8649	14.7467	0.2024
4.600	72.7872	14.7528	0.2027
4.605	72.8570	14.7681	0.2027
4.610	72.8198	14.7717	0.2029
4.615	72.7590	14.7207	0.2023
4.620	72.7942	14.7574	0.2027
4.625	72.7328	14.7467	0.2028
4.630	72.7613	14.8283	0.2038
4.635	72.7359	14.8195	0.2037
4.640	72.7505	14.8263	0.2038
4.645	72.7549	14.8286	0.2038
4.650	72.7549	14.8533	0.2042
4.655	72.6884	14.8974	0.2049
4.660	72.6988	14.8427	0.2042
4.665	72.7083	14.9042	0.2050
4.670	72.6645	14.8974	0.2050
4.675	72.6930	14.9293	0.2054
4.680	72.6537	14.9180	0.2053
4.685	72.6824	14.9714	0.2060

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.690	72.6790	14.9690	0.2060
4.695	72.6367	14.9596	0.2060
4.700	72.6500	14.9836	0.2062
4.705	72.6703	14.9760	0.2061
4.710	72.6197	15.0205	0.2068
4.715	72.6331	15.0202	0.2068
4.720	72.6410	15.0055	0.2066
4.725	72.6482	15.0675	0.2074
4.730	72.5753	15.0588	0.2075
4.735	72.5889	15.1248	0.2084
4.740	72.6043	15.1703	0.2089
4.745	72.6540	15.1341	0.2083
4.750	72.5908	15.1084	0.2081
4.755	72.5938	15.1643	0.2089
4.760	72.6355	15.1504	0.2086
4.765	72.5977	15.1607	0.2088
4.770	72.6055	15.1885	0.2092
4.775	72.6051	15.1703	0.2089
4.780	72.5617	15.2216	0.2098
4.785	72.5402	15.2107	0.2097
4.790	72.5545	15.2084	0.2096
4.795	72.5174	15.2552	0.2104
4.800	72.5246	15.2755	0.2106
4.805	72.5052	15.2655	0.2105
4.810	72.4849	15.2542	0.2104
4.815	72.4653	15.3592	0.2120
4.820	72.4847	15.3344	0.2116
4.825	72.4452	15.3698	0.2122
4.830	72.4587	15.4308	0.2130
4.835	72.4677	15.3975	0.2125
4.840	72.4336	15.4181	0.2129
4.845	72.4400	15.3869	0.2124
4.850	72.4992	15.4351	0.2129
4.855	72.4814	15.4309	0.2129
4.860	72.4692	15.4899	0.2137
4.865	72.3242	15.4515	0.2136
4.870	72.4204	15.4987	0.2140
4.875	72.4273	15.5371	0.2145
4.880	72.4720	15.5610	0.2147
4.885	72.3302	15.5476	0.2150
4.890	72.4099	15.6036	0.2155
4.895	72.3935	15.6267	0.2159
4.900	72.3871	15.6261	0.2159
4.905	72.3638	15.6482	0.2162
4.910	72.3361	15.6369	0.2162
4.915	72.3371	15.6365	0.2162

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.920	72.3290	15.6646	0.2166
4.925	72.2953	15.6758	0.2168
4.930	72.3071	15.6357	0.2162
4.935	72.3178	15.6521	0.2164
4.940	72.3120	15.6594	0.2166
4.945	72.3251	15.6754	0.2167
4.950	72.3314	15.6633	0.2165
4.955	72.3197	15.6752	0.2167
4.960	72.3280	15.7004	0.2171
4.965	72.3325	15.7247	0.2174
4.970	72.3564	15.7226	0.2173
4.975	72.3396	15.7248	0.2174
4.980	72.3464	15.7394	0.2176
4.985	72.3833	15.7848	0.2181
4.990	72.3689	15.7951	0.2183
4.995	72.3645	15.8175	0.2186
5.000	72.3840	15.8551	0.2190
5.005	72.3645	15.8579	0.2191
5.010	72.3424	15.8886	0.2196
5.015	72.3570	15.8948	0.2197
5.020	72.3743	15.9586	0.2205
5.025	72.3670	15.9969	0.2211
5.030	72.3193	16.0156	0.2215
5.035	72.2871	16.0867	0.2225
5.040	72.2980	16.1272	0.2231
5.045	72.2667	16.1426	0.2234
5.050	72.2016	16.1621	0.2238
5.055	72.1471	16.1342	0.2236
5.060	72.1374	16.1894	0.2244
5.065	72.0299	16.1543	0.2243
5.070	72.0318	16.1399	0.2241
5.075	71.9980	16.1418	0.2242
5.080	71.9980	16.1488	0.2243
5.085	71.9553	16.1724	0.2248
5.090	71.9986	16.1214	0.2239
5.095	72.0498	16.2014	0.2249
5.100	71.9835	16.1285	0.2241
5.105	71.9864	16.0908	0.2235
5.110	72.0458	16.1191	0.2237
5.115	72.0275	16.1306	0.2240
5.120	72.0482	16.1100	0.2236
5.125	72.0256	16.1736	0.2246
5.130	72.0619	16.1790	0.2245
5.135	72.0414	16.1844	0.2247
5.140	72.0423	16.1970	0.2248
5.145	71.9855	16.2391	0.2256

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.150	72.0931	16.2652	0.2256
5.155	72.0173	16.2866	0.2261
5.160	72.0068	16.2909	0.2262
5.165	71.9972	16.2949	0.2263
5.170	72.0043	16.3231	0.2267
5.175	72.0291	16.3548	0.2271
5.180	71.9808	16.3215	0.2267
5.185	72.0518	16.3758	0.2273
5.190	71.9824	16.3608	0.2273
5.195	71.9820	16.4058	0.2279
5.200	71.9344	16.4407	0.2286
5.205	71.9076	16.4115	0.2282
5.210	71.9457	16.4122	0.2281
5.215	71.9208	16.4822	0.2292
5.220	71.9306	16.4721	0.2290
5.225	71.8329	16.4754	0.2294
5.230	71.8828	16.5051	0.2296
5.235	71.8896	16.4955	0.2295
5.240	71.8790	16.5313	0.2300
5.245	71.8782	16.5350	0.2300
5.250	71.8803	16.5645	0.2304
5.255	71.8252	16.5895	0.2310
5.260	71.8575	16.6031	0.2311
5.265	71.8482	16.5850	0.2308
5.270	71.8171	16.6365	0.2317
5.275	71.8242	16.6187	0.2314
5.280	71.8409	16.6397	0.2316
5.285	71.8324	16.6387	0.2316
5.290	71.8198	16.6516	0.2319
5.295	71.7695	16.6336	0.2318
5.300	71.8006	16.6785	0.2323
5.305	71.7911	16.7118	0.2328
5.310	71.8058	16.7166	0.2328
5.315	71.7770	16.6991	0.2327
5.320	71.7327	16.7656	0.2337
5.325	71.7201	16.7109	0.2330
5.330	71.7438	16.7563	0.2336
5.335	71.7618	16.7874	0.2339
5.340	71.7135	16.7691	0.2338
5.345	71.6545	16.7653	0.2340
5.350	71.6905	16.7834	0.2341
5.355	71.6731	16.8101	0.2345
5.360	71.6963	16.8572	0.2351
5.365	71.6452	16.9023	0.2359
5.370	71.6715	16.8867	0.2356
5.375	71.6477	16.9100	0.2360

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.380	71.6412	16.9613	0.2368
5.385	71.6664	16.8494	0.2351
5.390	71.6752	16.9250	0.2361
5.395	71.6982	16.9709	0.2367
5.400	71.6322	16.9761	0.2370
5.405	71.6209	16.9496	0.2367
5.410	71.6394	16.9587	0.2367
5.415	71.6201	17.0476	0.2380
5.420	71.5940	17.0043	0.2375
5.425	71.5404	17.0206	0.2379
5.430	71.5805	17.0722	0.2385
5.435	71.5992	17.0852	0.2386
5.440	71.5877	17.0932	0.2388
5.445	71.5654	17.1181	0.2392
5.450	71.5656	17.0586	0.2384
5.455	71.5790	17.0944	0.2388
5.460	71.5104	17.1404	0.2397
5.465	71.5061	17.1495	0.2398
5.470	71.5499	17.1804	0.2401
5.475	71.5107	17.1921	0.2404
5.480	71.5152	17.2177	0.2408
5.485	71.5134	17.2626	0.2414
5.490	71.5466	17.2503	0.2411
5.495	71.5322	17.2764	0.2415
5.500	71.5233	17.2687	0.2414
5.505	71.5528	17.2572	0.2412
5.510	71.4856	17.3135	0.2422
5.515	71.4861	17.3646	0.2429
5.520	71.5000	17.3698	0.2429
5.525	71.4652	17.3188	0.2423
5.530	71.4701	17.3701	0.2430
5.535	71.4758	17.3880	0.2433
5.540	71.4558	17.3965	0.2435
5.545	71.4908	17.3979	0.2434
5.550	71.4595	17.4301	0.2439
5.555	71.4429	17.4357	0.2441
5.560	71.4448	17.4820	0.2447
5.565	71.4466	17.4695	0.2445
5.570	71.4255	17.4883	0.2448
5.575	71.4295	17.5176	0.2452
5.580	71.4349	17.5231	0.2453
5.585	71.4087	17.5340	0.2455
5.590	71.4178	17.5507	0.2457
5.595	71.4189	17.5801	0.2462
5.600	71.4144	17.5788	0.2462
5.605	71.3939	17.5888	0.2464

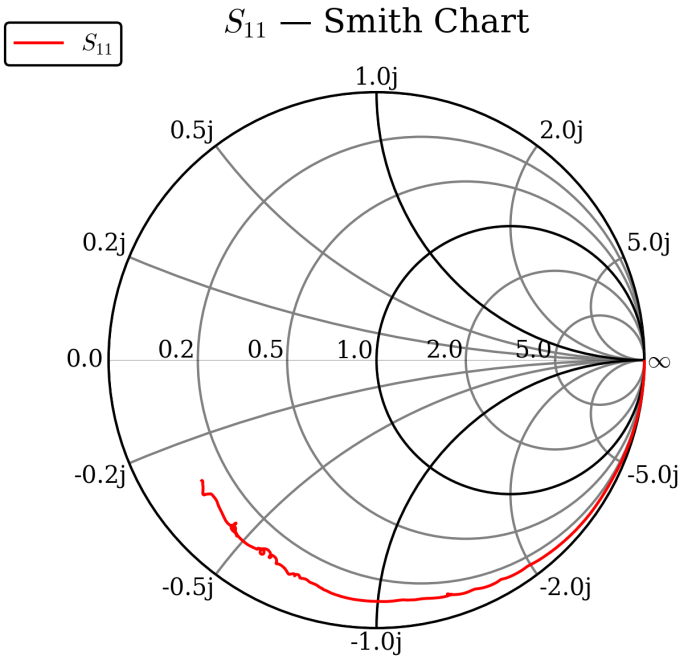
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.610	71.3864	17.5985	0.2465
5.615	71.4170	17.6141	0.2466
5.620	71.3593	17.6102	0.2468
5.625	71.3800	17.6329	0.2470
5.630	71.3803	17.6323	0.2470
5.635	71.3752	17.6406	0.2472
5.640	71.3781	17.6724	0.2476
5.645	71.3672	17.6755	0.2477
5.650	71.3593	17.6686	0.2476
5.655	71.3223	17.6799	0.2479
5.660	71.3162	17.6586	0.2476
5.665	71.3192	17.6954	0.2481
5.670	71.3053	17.7499	0.2489
5.675	71.2540	17.7167	0.2486
5.680	71.2695	17.7337	0.2488
5.685	71.2794	17.7191	0.2486
5.690	71.2909	17.7666	0.2492
5.695	71.2255	17.8078	0.2500
5.700	71.2877	17.7557	0.2491
5.705	71.1899	17.8232	0.2504
5.710	71.2393	17.7899	0.2497
5.715	71.1690	17.8417	0.2507
5.720	71.2055	17.9097	0.2515
5.725	71.2183	17.8783	0.2510
5.730	71.2474	17.8744	0.2509
5.735	71.2065	17.9061	0.2515
5.740	71.1738	17.8094	0.2502
5.745	71.1615	17.8962	0.2515
5.750	71.0965	17.8951	0.2517
5.755	71.1249	17.9112	0.2518
5.760	71.1734	17.9164	0.2517
5.765	71.0674	18.0078	0.2534
5.770	71.1144	18.0147	0.2533
5.775	71.0212	17.9080	0.2521
5.780	71.1190	17.9802	0.2528
5.785	71.0750	17.9569	0.2526
5.790	71.0490	18.0541	0.2541
5.795	71.0423	18.0405	0.2539
5.800	71.0424	18.0368	0.2539
5.805	71.0128	18.1027	0.2549
5.810	71.0076	18.1017	0.2549
5.815	70.9697	18.1227	0.2554
5.820	70.9588	18.1073	0.2552
5.825	71.0396	18.1526	0.2555
5.830	70.9715	18.1386	0.2556
5.835	70.9194	18.1653	0.2561

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.840	70.9159	18.1476	0.2559
5.845	70.9323	18.1749	0.2562
5.850	70.9033	18.2126	0.2569
5.855	70.9289	18.2074	0.2567
5.860	70.8840	18.2390	0.2573
5.865	70.9526	18.2489	0.2572
5.870	70.8393	18.2597	0.2578
5.875	70.8582	18.2803	0.2580
5.880	70.8594	18.2872	0.2581
5.885	70.8499	18.2971	0.2583
5.890	70.8355	18.2828	0.2581
5.895	70.8453	18.3107	0.2585
5.900	70.7973	18.3138	0.2587
5.905	70.8031	18.3869	0.2597
5.910	70.7902	18.3600	0.2594
5.915	70.7969	18.3864	0.2597
5.920	70.8002	18.3723	0.2595
5.925	70.7433	18.4405	0.2607
5.930	70.7215	18.3881	0.2600
5.935	70.7345	18.4045	0.2602
5.940	70.7147	18.4643	0.2611
5.945	70.7563	18.4386	0.2606
5.950	70.7647	18.4709	0.2610
5.955	70.6894	18.4793	0.2614
5.960	70.7317	18.5116	0.2617
5.965	70.7944	18.5447	0.2620
5.970	70.7169	18.4940	0.2615
5.975	70.6369	18.5262	0.2623
5.980	70.7443	18.5921	0.2628
5.985	70.6433	18.5465	0.2625
5.990	70.6971	18.5923	0.2630
5.995	70.6390	18.6058	0.2634
6.000	70.6642	18.5890	0.2631

1.2 S_{11} — Smith Chart



Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
0.005	0.9999	-0.0048	-0.00
0.010	0.9998	-0.0097	-0.00
0.015	0.9997	-0.0145	-0.00
0.020	0.9996	-0.0194	-0.00
0.025	0.9996	-0.0242	-0.00
0.030	0.9994	-0.0291	-0.00
0.035	0.9991	-0.0338	-0.00
0.040	0.9990	-0.0387	-0.00
0.045	0.9988	-0.0434	-0.00
0.050	0.9985	-0.0483	-0.00
0.055	0.9983	-0.0531	-0.00
0.060	0.9979	-0.0578	-0.00
0.065	0.9977	-0.0628	-0.00
0.070	0.9973	-0.0675	-0.00
0.075	0.9968	-0.0723	-0.00
0.080	0.9965	-0.0770	-0.00
0.085	0.9961	-0.0818	-0.00
0.090	0.9954	-0.0866	-0.01
0.095	0.9949	-0.0912	-0.01
0.100	0.9946	-0.0961	-0.01
0.105	0.9940	-0.1010	-0.01
0.110	0.9934	-0.1056	-0.01
0.115	0.9929	-0.1105	-0.01

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	0.9923	-0.1152	-0.01
0.125	0.9916	-0.1199	-0.01
0.130	0.9910	-0.1246	-0.01
0.135	0.9903	-0.1293	-0.01
0.140	0.9896	-0.1341	-0.01
0.145	0.9889	-0.1389	-0.01
0.150	0.9880	-0.1437	-0.01
0.155	0.9872	-0.1481	-0.02
0.160	0.9864	-0.1528	-0.02
0.165	0.9856	-0.1575	-0.02
0.170	0.9849	-0.1622	-0.02
0.175	0.9839	-0.1669	-0.02
0.180	0.9831	-0.1715	-0.02
0.185	0.9821	-0.1763	-0.02
0.190	0.9810	-0.1810	-0.02
0.195	0.9800	-0.1858	-0.02
0.200	0.9791	-0.1904	-0.02
0.205	0.9780	-0.1951	-0.02
0.210	0.9768	-0.1998	-0.03
0.215	0.9757	-0.2046	-0.03
0.220	0.9745	-0.2091	-0.03
0.225	0.9734	-0.2136	-0.03
0.230	0.9722	-0.2182	-0.03
0.235	0.9710	-0.2229	-0.03
0.240	0.9698	-0.2275	-0.03
0.245	0.9686	-0.2321	-0.03
0.250	0.9674	-0.2368	-0.04
0.255	0.9659	-0.2412	-0.04
0.260	0.9645	-0.2458	-0.04
0.265	0.9632	-0.2504	-0.04
0.270	0.9618	-0.2548	-0.04
0.275	0.9605	-0.2592	-0.05
0.280	0.9590	-0.2638	-0.05
0.285	0.9577	-0.2683	-0.05
0.290	0.9562	-0.2729	-0.05
0.295	0.9548	-0.2773	-0.05
0.300	0.9533	-0.2818	-0.05
0.305	0.9517	-0.2864	-0.05
0.310	0.9501	-0.2909	-0.06
0.315	0.9484	-0.2952	-0.06
0.320	0.9468	-0.2995	-0.06
0.325	0.9453	-0.3039	-0.06
0.330	0.9436	-0.3084	-0.06
0.335	0.9420	-0.3126	-0.07
0.340	0.9403	-0.3171	-0.07
0.345	0.9387	-0.3214	-0.07

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.350	0.9369	-0.3257	-0.07
0.355	0.9353	-0.3302	-0.07
0.360	0.9334	-0.3344	-0.07
0.365	0.9316	-0.3388	-0.08
0.370	0.9297	-0.3431	-0.08
0.375	0.9279	-0.3474	-0.08
0.380	0.9259	-0.3515	-0.08
0.385	0.9240	-0.3557	-0.09
0.390	0.9224	-0.3601	-0.09
0.395	0.9204	-0.3641	-0.09
0.400	0.9187	-0.3683	-0.09
0.405	0.9167	-0.3724	-0.09
0.410	0.9149	-0.3765	-0.09
0.415	0.9129	-0.3806	-0.10
0.420	0.9111	-0.3846	-0.10
0.425	0.9090	-0.3888	-0.10
0.430	0.9071	-0.3928	-0.10
0.435	0.9052	-0.3967	-0.10
0.440	0.9031	-0.4008	-0.10
0.445	0.9013	-0.4049	-0.10
0.450	0.8993	-0.4088	-0.11
0.455	0.8971	-0.4127	-0.11
0.460	0.8952	-0.4166	-0.11
0.465	0.8929	-0.4204	-0.11
0.470	0.8910	-0.4243	-0.11
0.475	0.8889	-0.4282	-0.12
0.480	0.8869	-0.4319	-0.12
0.485	0.8847	-0.4359	-0.12
0.490	0.8827	-0.4395	-0.12
0.495	0.8806	-0.4434	-0.12
0.500	0.8786	-0.4472	-0.12
0.505	0.8764	-0.4510	-0.13
0.510	0.8743	-0.4547	-0.13
0.515	0.8723	-0.4584	-0.13
0.520	0.8702	-0.4620	-0.13
0.525	0.8682	-0.4656	-0.13
0.530	0.8662	-0.4693	-0.13
0.535	0.8641	-0.4730	-0.13
0.540	0.8619	-0.4766	-0.13
0.545	0.8598	-0.4803	-0.13
0.550	0.8575	-0.4838	-0.13
0.555	0.8552	-0.4875	-0.14
0.560	0.8531	-0.4910	-0.14
0.565	0.8509	-0.4946	-0.14
0.570	0.8488	-0.4982	-0.14
0.575	0.8466	-0.5018	-0.14

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.8443	-0.5053	-0.14
0.585	0.8421	-0.5087	-0.14
0.590	0.8398	-0.5123	-0.14
0.595	0.8377	-0.5156	-0.14
0.600	0.8354	-0.5190	-0.14
0.605	0.8331	-0.5225	-0.15
0.610	0.8309	-0.5258	-0.15
0.615	0.8286	-0.5291	-0.15
0.620	0.8263	-0.5327	-0.15
0.625	0.8239	-0.5360	-0.15
0.630	0.8217	-0.5394	-0.15
0.635	0.8194	-0.5428	-0.15
0.640	0.8171	-0.5461	-0.15
0.645	0.8147	-0.5494	-0.15
0.650	0.8124	-0.5525	-0.15
0.655	0.8102	-0.5559	-0.15
0.660	0.8078	-0.5592	-0.15
0.665	0.8054	-0.5623	-0.16
0.670	0.8031	-0.5655	-0.16
0.675	0.8006	-0.5687	-0.16
0.680	0.7980	-0.5719	-0.16
0.685	0.7956	-0.5750	-0.16
0.690	0.7932	-0.5780	-0.16
0.695	0.7906	-0.5811	-0.17
0.700	0.7880	-0.5842	-0.17
0.705	0.7855	-0.5875	-0.17
0.710	0.7829	-0.5906	-0.17
0.715	0.7805	-0.5937	-0.17
0.720	0.7778	-0.5967	-0.17
0.725	0.7752	-0.5998	-0.17
0.730	0.7725	-0.6027	-0.18
0.735	0.7699	-0.6059	-0.18
0.740	0.7674	-0.6088	-0.18
0.745	0.7646	-0.6117	-0.18
0.750	0.7621	-0.6147	-0.18
0.755	0.7593	-0.6175	-0.19
0.760	0.7566	-0.6203	-0.19
0.765	0.7540	-0.6233	-0.19
0.770	0.7513	-0.6263	-0.19
0.775	0.7485	-0.6292	-0.19
0.780	0.7458	-0.6323	-0.20
0.785	0.7429	-0.6353	-0.20
0.790	0.7400	-0.6381	-0.20
0.795	0.7372	-0.6410	-0.20
0.800	0.7343	-0.6437	-0.21
0.805	0.7314	-0.6466	-0.21

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.810	0.7286	-0.6493	-0.21
0.815	0.7255	-0.6522	-0.21
0.820	0.7226	-0.6549	-0.22
0.825	0.7197	-0.6577	-0.22
0.830	0.7169	-0.6606	-0.22
0.835	0.7137	-0.6633	-0.23
0.840	0.7107	-0.6660	-0.23
0.845	0.7079	-0.6688	-0.23
0.850	0.7047	-0.6715	-0.23
0.855	0.7018	-0.6743	-0.24
0.860	0.6987	-0.6769	-0.24
0.865	0.6955	-0.6797	-0.24
0.870	0.6924	-0.6823	-0.25
0.875	0.6893	-0.6850	-0.25
0.880	0.6861	-0.6875	-0.25
0.885	0.6829	-0.6901	-0.26
0.890	0.6799	-0.6928	-0.26
0.895	0.6765	-0.6954	-0.26
0.900	0.6734	-0.6980	-0.27
0.905	0.6703	-0.7007	-0.27
0.910	0.6670	-0.7031	-0.27
0.915	0.6636	-0.7057	-0.28
0.920	0.6604	-0.7082	-0.28
0.925	0.6572	-0.7107	-0.28
0.930	0.6538	-0.7133	-0.29
0.935	0.6505	-0.7157	-0.29
0.940	0.6470	-0.7182	-0.29
0.945	0.6435	-0.7206	-0.30
0.950	0.6402	-0.7230	-0.30
0.955	0.6368	-0.7255	-0.31
0.960	0.6334	-0.7277	-0.31
0.965	0.6300	-0.7301	-0.31
0.970	0.6268	-0.7324	-0.32
0.975	0.6234	-0.7346	-0.32
0.980	0.6202	-0.7370	-0.33
0.985	0.6167	-0.7391	-0.33
0.990	0.6133	-0.7413	-0.34
0.995	0.6101	-0.7435	-0.34
1.000	0.6065	-0.7457	-0.34
1.005	0.6031	-0.7479	-0.35
1.010	0.5995	-0.7499	-0.35
1.015	0.5960	-0.7518	-0.36
1.020	0.5925	-0.7539	-0.37
1.025	0.5892	-0.7557	-0.37
1.030	0.5856	-0.7577	-0.38
1.035	0.5822	-0.7595	-0.38

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.040	0.5789	-0.7613	-0.39
1.045	0.5757	-0.7629	-0.39
1.050	0.5725	-0.7644	-0.40
1.055	0.5694	-0.7659	-0.41
1.060	0.5666	-0.7677	-0.41
1.065	0.5637	-0.7692	-0.41
1.070	0.5611	-0.7710	-0.41
1.075	0.5583	-0.7731	-0.41
1.080	0.5556	-0.7750	-0.41
1.085	0.5529	-0.7771	-0.41
1.090	0.5500	-0.7793	-0.41
1.095	0.5469	-0.7815	-0.41
1.100	0.5437	-0.7837	-0.41
1.105	0.5405	-0.7857	-0.41
1.110	0.5373	-0.7876	-0.41
1.115	0.5340	-0.7896	-0.42
1.120	0.5308	-0.7915	-0.42
1.125	0.5274	-0.7934	-0.42
1.130	0.5243	-0.7951	-0.42
1.135	0.5209	-0.7968	-0.43
1.140	0.5177	-0.7984	-0.43
1.145	0.5146	-0.8001	-0.43
1.150	0.5114	-0.8018	-0.44
1.155	0.5081	-0.8033	-0.44
1.160	0.5049	-0.8049	-0.44
1.165	0.5018	-0.8064	-0.45
1.170	0.4985	-0.8077	-0.45
1.175	0.4954	-0.8091	-0.46
1.180	0.4922	-0.8105	-0.46
1.185	0.4891	-0.8115	-0.47
1.190	0.4860	-0.8127	-0.47
1.195	0.4832	-0.8137	-0.48
1.200	0.4802	-0.8145	-0.49
1.205	0.4771	-0.8151	-0.50
1.210	0.4744	-0.8158	-0.50
1.215	0.4718	-0.8164	-0.51
1.220	0.4694	-0.8169	-0.52
1.225	0.4671	-0.8175	-0.52
1.230	0.4649	-0.8184	-0.53
1.235	0.4630	-0.8194	-0.53
1.240	0.4611	-0.8207	-0.52
1.245	0.4591	-0.8222	-0.52
1.250	0.4570	-0.8237	-0.52
1.255	0.4547	-0.8252	-0.52
1.260	0.4524	-0.8267	-0.52
1.265	0.4498	-0.8285	-0.51

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.4473	-0.8300	-0.51
1.275	0.4446	-0.8315	-0.51
1.280	0.4420	-0.8329	-0.51
1.285	0.4389	-0.8344	-0.51
1.290	0.4362	-0.8356	-0.51
1.295	0.4333	-0.8371	-0.51
1.300	0.4304	-0.8383	-0.52
1.305	0.4277	-0.8396	-0.52
1.310	0.4249	-0.8408	-0.52
1.315	0.4221	-0.8419	-0.52
1.320	0.4192	-0.8431	-0.52
1.325	0.4163	-0.8441	-0.53
1.330	0.4135	-0.8449	-0.53
1.335	0.4105	-0.8459	-0.54
1.340	0.4077	-0.8469	-0.54
1.345	0.4047	-0.8476	-0.54
1.350	0.4017	-0.8485	-0.55
1.355	0.3988	-0.8494	-0.55
1.360	0.3958	-0.8500	-0.56
1.365	0.3927	-0.8508	-0.56
1.370	0.3898	-0.8514	-0.57
1.375	0.3868	-0.8518	-0.58
1.380	0.3839	-0.8521	-0.59
1.385	0.3810	-0.8525	-0.59
1.390	0.3782	-0.8529	-0.60
1.395	0.3754	-0.8533	-0.61
1.400	0.3728	-0.8536	-0.62
1.405	0.3702	-0.8539	-0.62
1.410	0.3677	-0.8545	-0.63
1.415	0.3652	-0.8550	-0.63
1.420	0.3628	-0.8560	-0.63
1.425	0.3604	-0.8568	-0.63
1.430	0.3578	-0.8579	-0.64
1.435	0.3553	-0.8589	-0.63
1.440	0.3525	-0.8604	-0.63
1.445	0.3496	-0.8617	-0.63
1.450	0.3467	-0.8629	-0.63
1.455	0.3437	-0.8644	-0.63
1.460	0.3402	-0.8658	-0.63
1.465	0.3369	-0.8668	-0.63
1.470	0.3334	-0.8679	-0.63
1.475	0.3298	-0.8691	-0.63
1.480	0.3262	-0.8702	-0.64
1.485	0.3223	-0.8710	-0.64
1.490	0.3186	-0.8720	-0.65
1.495	0.3147	-0.8727	-0.65

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.500	0.3109	-0.8735	-0.66
1.505	0.3068	-0.8739	-0.67
1.510	0.3028	-0.8743	-0.67
1.515	0.2988	-0.8747	-0.68
1.520	0.2947	-0.8751	-0.69
1.525	0.2908	-0.8752	-0.70
1.530	0.2870	-0.8751	-0.72
1.535	0.2828	-0.8749	-0.73
1.540	0.2789	-0.8748	-0.74
1.545	0.2750	-0.8745	-0.76
1.550	0.2715	-0.8739	-0.77
1.555	0.2679	-0.8734	-0.79
1.560	0.2644	-0.8725	-0.80
1.565	0.2807	-0.8746	-0.74
1.570	0.2775	-0.8757	-0.74
1.575	0.2739	-0.8770	-0.74
1.580	0.2706	-0.8780	-0.74
1.585	0.2670	-0.8791	-0.74
1.590	0.2633	-0.8803	-0.74
1.595	0.2598	-0.8814	-0.73
1.600	0.2561	-0.8822	-0.74
1.605	0.2526	-0.8833	-0.74
1.610	0.2490	-0.8839	-0.74
1.615	0.2451	-0.8846	-0.74
1.620	0.2410	-0.8853	-0.75
1.625	0.2375	-0.8858	-0.75
1.630	0.2335	-0.8864	-0.76
1.635	0.2298	-0.8867	-0.76
1.640	0.2260	-0.8872	-0.77
1.645	0.2222	-0.8875	-0.77
1.650	0.2183	-0.8878	-0.78
1.655	0.2147	-0.8880	-0.78
1.660	0.2111	-0.8883	-0.79
1.665	0.2075	-0.8885	-0.80
1.670	0.2042	-0.8885	-0.80
1.675	0.2007	-0.8887	-0.81
1.680	0.1975	-0.8889	-0.81
1.685	0.1943	-0.8888	-0.82
1.690	0.1912	-0.8890	-0.83
1.695	0.1883	-0.8891	-0.83
1.700	0.1853	-0.8891	-0.84
1.705	0.1822	-0.8893	-0.84
1.710	0.1792	-0.8896	-0.84
1.715	0.1762	-0.8898	-0.85
1.720	0.1731	-0.8902	-0.85
1.725	0.1701	-0.8904	-0.85

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.730	0.1670	-0.8909	-0.85
1.735	0.1642	-0.8915	-0.85
1.740	0.1611	-0.8917	-0.86
1.745	0.1582	-0.8922	-0.86
1.750	0.1553	-0.8927	-0.86
1.755	0.1524	-0.8931	-0.86
1.760	0.1496	-0.8935	-0.86
1.765	0.1469	-0.8939	-0.86
1.770	0.1444	-0.8942	-0.86
1.775	0.1416	-0.8946	-0.86
1.780	0.1391	-0.8946	-0.86
1.785	0.1362	-0.8946	-0.87
1.790	0.1334	-0.8946	-0.87
1.795	0.1305	-0.8948	-0.87
1.800	0.1279	-0.8948	-0.88
1.805	0.1251	-0.8951	-0.88
1.810	0.1227	-0.8952	-0.88
1.815	0.1203	-0.8953	-0.88
1.820	0.1178	-0.8954	-0.89
1.825	0.1156	-0.8954	-0.89
1.830	0.1133	-0.8953	-0.89
1.835	0.1110	-0.8952	-0.90
1.840	0.1088	-0.8953	-0.90
1.845	0.1067	-0.8949	-0.90
1.850	0.1045	-0.8948	-0.91
1.855	0.1023	-0.8948	-0.91
1.860	0.1001	-0.8946	-0.91
1.865	0.0982	-0.8945	-0.92
1.870	0.0960	-0.8948	-0.92
1.875	0.0940	-0.8951	-0.91
1.880	0.0920	-0.8953	-0.91
1.885	0.0904	-0.8955	-0.91
1.890	0.0888	-0.8957	-0.91
1.895	0.0871	-0.8959	-0.91
1.900	0.0852	-0.8958	-0.92
1.905	0.0834	-0.8960	-0.92
1.910	0.0814	-0.8962	-0.92
1.915	0.0794	-0.8965	-0.91
1.920	0.0777	-0.8970	-0.91
1.925	0.0760	-0.8973	-0.91
1.930	0.0744	-0.8973	-0.91
1.935	0.0725	-0.8977	-0.91
1.940	0.0707	-0.8979	-0.91
1.945	0.0690	-0.8983	-0.91
1.950	0.0673	-0.8984	-0.91
1.955	0.0654	-0.8986	-0.91

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.960	0.0639	-0.8987	-0.91
1.965	0.0621	-0.8989	-0.91
1.970	0.0604	-0.8991	-0.90
1.975	0.0586	-0.8994	-0.90
1.980	0.0570	-0.8994	-0.90
1.985	0.0553	-0.8996	-0.90
1.990	0.0535	-0.8998	-0.90
1.995	0.0517	-0.9000	-0.90
2.000	0.0498	-0.9001	-0.90
2.005	0.0479	-0.9001	-0.90
2.010	0.0461	-0.9004	-0.90
2.015	0.0443	-0.9003	-0.90
2.020	0.0423	-0.9005	-0.90
2.025	0.0404	-0.9007	-0.90
2.030	0.0385	-0.9008	-0.90
2.035	0.0364	-0.9008	-0.90
2.040	0.0344	-0.9008	-0.90
2.045	0.0326	-0.9008	-0.90
2.050	0.0304	-0.9007	-0.90
2.055	0.0286	-0.9009	-0.90
2.060	0.0263	-0.9008	-0.90
2.065	0.0242	-0.9010	-0.90
2.070	0.0218	-0.9008	-0.90
2.075	0.0198	-0.9008	-0.91
2.080	0.0175	-0.9007	-0.91
2.085	0.0151	-0.9005	-0.91
2.090	0.0127	-0.9005	-0.91
2.095	0.0103	-0.9004	-0.91
2.100	0.0079	-0.9004	-0.91
2.105	0.0054	-0.9005	-0.91
2.110	0.0028	-0.9002	-0.91
2.115	0.0003	-0.9003	-0.91
2.120	-0.0024	-0.9003	-0.91
2.125	-0.0049	-0.9002	-0.91
2.130	-0.0077	-0.9002	-0.91
2.135	-0.0105	-0.9000	-0.91
2.140	-0.0133	-0.8999	-0.91
2.145	-0.0160	-0.8998	-0.92
2.150	-0.0187	-0.8997	-0.92
2.155	-0.0217	-0.8997	-0.92
2.160	-0.0245	-0.8994	-0.92
2.165	-0.0273	-0.8992	-0.92
2.170	-0.0304	-0.8988	-0.92
2.175	-0.0333	-0.8988	-0.92
2.180	-0.0362	-0.8986	-0.92
2.185	-0.0391	-0.8983	-0.92

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.190	-0.0423	-0.8980	-0.92
2.195	-0.0453	-0.8976	-0.93
2.200	-0.0486	-0.8975	-0.93
2.205	-0.0516	-0.8971	-0.93
2.210	-0.0547	-0.8967	-0.93
2.215	-0.0576	-0.8962	-0.93
2.220	-0.0608	-0.8961	-0.93
2.225	-0.0640	-0.8956	-0.94
2.230	-0.0670	-0.8952	-0.94
2.235	-0.0703	-0.8948	-0.94
2.240	-0.0734	-0.8943	-0.94
2.245	-0.0765	-0.8939	-0.94
2.250	-0.0795	-0.8935	-0.94
2.255	-0.0825	-0.8931	-0.94
2.260	-0.0856	-0.8926	-0.95
2.265	-0.0885	-0.8920	-0.95
2.270	-0.0915	-0.8915	-0.95
2.275	-0.0946	-0.8908	-0.96
2.280	-0.0974	-0.8902	-0.96
2.285	-0.1003	-0.8896	-0.96
2.290	-0.1033	-0.8890	-0.96
2.295	-0.1061	-0.8884	-0.97
2.300	-0.1090	-0.8877	-0.97
2.305	-0.1120	-0.8871	-0.97
2.310	-0.1147	-0.8863	-0.98
2.315	-0.1175	-0.8856	-0.98
2.320	-0.1204	-0.8849	-0.98
2.325	-0.1232	-0.8845	-0.98
2.330	-0.1259	-0.8837	-0.99
2.335	-0.1285	-0.8829	-0.99
2.340	-0.1312	-0.8821	-0.99
2.345	-0.1339	-0.8813	-1.00
2.350	-0.1363	-0.8804	-1.00
2.355	-0.1388	-0.8795	-1.01
2.360	-0.1411	-0.8789	-1.01
2.365	-0.1434	-0.8782	-1.01
2.370	-0.1456	-0.8775	-1.02
2.375	-0.1477	-0.8767	-1.02
2.380	-0.1501	-0.8760	-1.02
2.385	-0.1521	-0.8752	-1.03
2.390	-0.1542	-0.8746	-1.03
2.395	-0.1566	-0.8739	-1.03
2.400	-0.1585	-0.8733	-1.04
2.405	-0.1603	-0.8724	-1.04
2.410	-0.1623	-0.8718	-1.04
2.415	-0.1642	-0.8712	-1.05

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.420	-0.1658	-0.8706	-1.05
2.425	-0.1675	-0.8696	-1.06
2.430	-0.1691	-0.8691	-1.06
2.435	-0.1708	-0.8683	-1.06
2.440	-0.1724	-0.8677	-1.06
2.445	-0.1740	-0.8670	-1.07
2.450	-0.1754	-0.8662	-1.07
2.455	-0.1765	-0.8656	-1.08
2.460	-0.1778	-0.8649	-1.08
2.465	-0.1792	-0.8642	-1.08
2.470	-0.1803	-0.8634	-1.09
2.475	-0.1816	-0.8631	-1.09
2.480	-0.1830	-0.8623	-1.09
2.485	-0.1842	-0.8618	-1.10
2.490	-0.1853	-0.8611	-1.10
2.495	-0.1866	-0.8606	-1.10
2.500	-0.1877	-0.8601	-1.11
2.505	-0.1889	-0.8597	-1.11
2.510	-0.1899	-0.8593	-1.11
2.515	-0.1910	-0.8586	-1.11
2.520	-0.1921	-0.8584	-1.11
2.525	-0.1932	-0.8580	-1.12
2.530	-0.1944	-0.8578	-1.11
2.535	-0.1954	-0.8573	-1.12
2.540	-0.1964	-0.8570	-1.12
2.545	-0.1974	-0.8566	-1.12
2.550	-0.1982	-0.8562	-1.12
2.555	-0.1988	-0.8557	-1.13
2.560	-0.1996	-0.8551	-1.13
2.565	-0.2001	-0.8545	-1.13
2.570	-0.2005	-0.8539	-1.14
2.575	-0.2007	-0.8533	-1.14
2.580	-0.2011	-0.8528	-1.15
2.585	-0.2013	-0.8528	-1.15
2.590	-0.2016	-0.8525	-1.15
2.595	-0.2021	-0.8526	-1.15
2.600	-0.2026	-0.8522	-1.15
2.605	-0.2030	-0.8522	-1.15
2.610	-0.2034	-0.8520	-1.15
2.615	-0.2039	-0.8517	-1.15
2.620	-0.2045	-0.8515	-1.15
2.625	-0.2049	-0.8513	-1.15
2.630	-0.2054	-0.8510	-1.16
2.635	-0.2059	-0.8509	-1.16
2.640	-0.2065	-0.8504	-1.16
2.645	-0.2071	-0.8501	-1.16

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.650	-0.2076	-0.8499	-1.16
2.655	-0.2081	-0.8492	-1.17
2.660	-0.2090	-0.8490	-1.17
2.665	-0.2096	-0.8486	-1.17
2.670	-0.2101	-0.8482	-1.17
2.675	-0.2108	-0.8480	-1.17
2.680	-0.2112	-0.8476	-1.17
2.685	-0.2116	-0.8475	-1.17
2.690	-0.2120	-0.8472	-1.18
2.695	-0.2125	-0.8472	-1.18
2.700	-0.2128	-0.8470	-1.18
2.705	-0.2131	-0.8469	-1.18
2.710	-0.2136	-0.8467	-1.18
2.715	-0.2140	-0.8463	-1.18
2.720	-0.2147	-0.8460	-1.18
2.725	-0.2150	-0.8453	-1.19
2.730	-0.2158	-0.8448	-1.19
2.735	-0.2166	-0.8444	-1.19
2.740	-0.2173	-0.8434	-1.20
2.745	-0.2183	-0.8430	-1.20
2.750	-0.2192	-0.8422	-1.21
2.755	-0.2204	-0.8415	-1.21
2.760	-0.2215	-0.8409	-1.21
2.765	-0.2226	-0.8400	-1.22
2.770	-0.2238	-0.8394	-1.22
2.775	-0.2251	-0.8386	-1.23
2.780	-0.2264	-0.8378	-1.23
2.785	-0.2279	-0.8371	-1.23
2.790	-0.2291	-0.8367	-1.23
2.795	-0.2308	-0.8360	-1.24
2.800	-0.2320	-0.8355	-1.24
2.805	-0.2335	-0.8348	-1.24
2.810	-0.2350	-0.8342	-1.24
2.815	-0.2364	-0.8335	-1.25
2.820	-0.2377	-0.8331	-1.25
2.825	-0.2390	-0.8325	-1.25
2.830	-0.2404	-0.8321	-1.25
2.835	-0.2418	-0.8315	-1.25
2.840	-0.2432	-0.8311	-1.25
2.845	-0.2448	-0.8305	-1.25
2.850	-0.2459	-0.8298	-1.26
2.855	-0.2471	-0.8291	-1.26
2.860	-0.2485	-0.8287	-1.26
2.865	-0.2496	-0.8281	-1.26
2.870	-0.2507	-0.8274	-1.26
2.875	-0.2516	-0.8265	-1.27

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.880	-0.2525	-0.8255	-1.28
2.885	-0.2536	-0.8244	-1.28
2.890	-0.2551	-0.8233	-1.29
2.895	-0.2567	-0.8219	-1.30
2.900	-0.2581	-0.8207	-1.31
2.905	-0.2597	-0.8198	-1.31
2.910	-0.2615	-0.8187	-1.32
2.915	-0.2635	-0.8179	-1.32
2.920	-0.2651	-0.8171	-1.32
2.925	-0.2667	-0.8166	-1.32
2.930	-0.2686	-0.8157	-1.32
2.935	-0.2701	-0.8155	-1.32
2.940	-0.2715	-0.8152	-1.32
2.945	-0.2730	-0.8149	-1.32
2.950	-0.2744	-0.8150	-1.31
2.955	-0.2756	-0.8148	-1.31
2.960	-0.2766	-0.8145	-1.31
2.965	-0.2775	-0.8146	-1.30
2.970	-0.2783	-0.8144	-1.30
2.975	-0.2791	-0.8147	-1.30
2.980	-0.2800	-0.8147	-1.29
2.985	-0.2805	-0.8146	-1.29
2.990	-0.2809	-0.8144	-1.29
2.995	-0.2812	-0.8141	-1.30
3.000	-0.2818	-0.8138	-1.30
3.005	-0.2854	-0.8126	-1.30
3.010	-0.2859	-0.8124	-1.30
3.015	-0.2861	-0.8125	-1.30
3.020	-0.2863	-0.8122	-1.30
3.025	-0.2864	-0.8124	-1.30
3.030	-0.2863	-0.8121	-1.30
3.035	-0.2863	-0.8117	-1.30
3.040	-0.2862	-0.8110	-1.31
3.045	-0.2865	-0.8103	-1.32
3.050	-0.2863	-0.8093	-1.33
3.055	-0.2862	-0.8086	-1.33
3.060	-0.2867	-0.8074	-1.34
3.065	-0.2874	-0.8063	-1.35
3.070	-0.2879	-0.8052	-1.36
3.075	-0.2886	-0.8041	-1.37
3.080	-0.2893	-0.8031	-1.37
3.085	-0.2905	-0.8018	-1.38
3.090	-0.2916	-0.8009	-1.39
3.095	-0.2929	-0.8007	-1.39
3.100	-0.2940	-0.8000	-1.39
3.105	-0.2956	-0.8000	-1.38

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.110	-0.2968	-0.7999	-1.38
3.115	-0.2983	-0.7996	-1.38
3.120	-0.2996	-0.7999	-1.37
3.125	-0.3011	-0.8001	-1.36
3.130	-0.3019	-0.8005	-1.36
3.135	-0.3029	-0.8011	-1.35
3.140	-0.3034	-0.8018	-1.34
3.145	-0.3044	-0.8026	-1.33
3.150	-0.3048	-0.8031	-1.32
3.155	-0.3050	-0.8036	-1.31
3.160	-0.3055	-0.8042	-1.31
3.165	-0.3055	-0.8045	-1.30
3.170	-0.3055	-0.8045	-1.30
3.175	-0.3057	-0.8052	-1.30
3.180	-0.3053	-0.8051	-1.30
3.185	-0.3052	-0.8048	-1.30
3.190	-0.3048	-0.8043	-1.31
3.195	-0.3048	-0.8035	-1.32
3.200	-0.3050	-0.8024	-1.33
3.205	-0.3051	-0.8015	-1.33
3.210	-0.3055	-0.8004	-1.34
3.215	-0.3059	-0.7993	-1.35
3.220	-0.3068	-0.7981	-1.36
3.225	-0.3084	-0.7975	-1.36
3.230	-0.3099	-0.7970	-1.36
3.235	-0.3111	-0.7970	-1.35
3.240	-0.3122	-0.7970	-1.35
3.245	-0.3135	-0.7973	-1.34
3.250	-0.3149	-0.7976	-1.34
3.255	-0.3160	-0.7978	-1.33
3.260	-0.3172	-0.7981	-1.32
3.265	-0.3183	-0.7984	-1.32
3.270	-0.3191	-0.7983	-1.31
3.275	-0.3199	-0.7985	-1.31
3.280	-0.3211	-0.7987	-1.30
3.285	-0.3224	-0.7988	-1.30
3.290	-0.3232	-0.7987	-1.29
3.295	-0.3239	-0.7989	-1.29
3.300	-0.3248	-0.7989	-1.29
3.305	-0.3257	-0.7990	-1.28
3.310	-0.3260	-0.7986	-1.28
3.315	-0.3268	-0.7985	-1.28
3.320	-0.3275	-0.7984	-1.28
3.325	-0.3279	-0.7983	-1.28
3.330	-0.3286	-0.7971	-1.29
3.335	-0.3295	-0.7968	-1.29

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.340	-0.3300	-0.7964	-1.29
3.345	-0.3306	-0.7955	-1.30
3.350	-0.3309	-0.7943	-1.31
3.355	-0.3318	-0.7933	-1.31
3.360	-0.3325	-0.7920	-1.32
3.365	-0.3332	-0.7912	-1.33
3.370	-0.3341	-0.7898	-1.33
3.375	-0.3349	-0.7886	-1.34
3.380	-0.3355	-0.7868	-1.36
3.385	-0.3366	-0.7851	-1.37
3.390	-0.3374	-0.7837	-1.38
3.395	-0.3389	-0.7821	-1.39
3.400	-0.3400	-0.7804	-1.40
3.405	-0.3413	-0.7789	-1.41
3.410	-0.3425	-0.7770	-1.42
3.415	-0.3443	-0.7754	-1.43
3.420	-0.3459	-0.7739	-1.44
3.425	-0.3472	-0.7724	-1.44
3.430	-0.3491	-0.7709	-1.45
3.435	-0.3509	-0.7694	-1.46
3.440	-0.3524	-0.7682	-1.46
3.445	-0.3546	-0.7666	-1.47
3.450	-0.3566	-0.7655	-1.47
3.455	-0.3585	-0.7640	-1.47
3.460	-0.3604	-0.7631	-1.47
3.465	-0.3622	-0.7621	-1.48
3.470	-0.3642	-0.7615	-1.47
3.475	-0.3659	-0.7609	-1.47
3.480	-0.3673	-0.7603	-1.47
3.485	-0.3693	-0.7599	-1.46
3.490	-0.3709	-0.7603	-1.45
3.495	-0.3719	-0.7598	-1.45
3.500	-0.3727	-0.7599	-1.45
3.505	-0.3735	-0.7598	-1.45
3.510	-0.3735	-0.7590	-1.45
3.515	-0.3740	-0.7586	-1.45
3.520	-0.3738	-0.7577	-1.46
3.525	-0.3746	-0.7566	-1.47
3.530	-0.3737	-0.7564	-1.48
3.535	-0.3729	-0.7540	-1.50
3.540	-0.3725	-0.7517	-1.53
3.545	-0.3727	-0.7494	-1.55
3.550	-0.3731	-0.7459	-1.58
3.555	-0.3739	-0.7429	-1.60
3.560	-0.3755	-0.7396	-1.62
3.565	-0.3770	-0.7368	-1.64

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.570	-0.3802	-0.7337	-1.66
3.575	-0.3835	-0.7313	-1.66
3.580	-0.3871	-0.7290	-1.67
3.585	-0.3907	-0.7283	-1.66
3.590	-0.3944	-0.7283	-1.64
3.595	-0.3977	-0.7297	-1.61
3.600	-0.3996	-0.7316	-1.58
3.605	-0.4007	-0.7337	-1.56
3.610	-0.4008	-0.7354	-1.54
3.615	-0.4004	-0.7366	-1.53
3.620	-0.3989	-0.7369	-1.54
3.625	-0.3971	-0.7374	-1.54
3.630	-0.3961	-0.7365	-1.55
3.635	-0.3946	-0.7355	-1.57
3.640	-0.3939	-0.7340	-1.59
3.645	-0.3937	-0.7319	-1.61
3.650	-0.3939	-0.7301	-1.62
3.655	-0.3942	-0.7283	-1.64
3.660	-0.3944	-0.7266	-1.65
3.665	-0.3951	-0.7247	-1.67
3.670	-0.3957	-0.7229	-1.68
3.675	-0.3957	-0.7213	-1.70
3.680	-0.3962	-0.7193	-1.71
3.685	-0.3973	-0.7173	-1.72
3.690	-0.3986	-0.7154	-1.73
3.695	-0.4002	-0.7132	-1.75
3.700	-0.4019	-0.7108	-1.76
3.705	-0.4046	-0.7091	-1.76
3.710	-0.4074	-0.7065	-1.77
3.715	-0.4111	-0.7053	-1.76
3.720	-0.4152	-0.7043	-1.75
3.725	-0.4203	-0.7037	-1.73
3.730	-0.4251	-0.7044	-1.70
3.735	-0.4285	-0.7067	-1.66
3.740	-0.4322	-0.7099	-1.61
3.745	-0.4335	-0.7129	-1.57
3.750	-0.4344	-0.7169	-1.53
3.755	-0.4337	-0.7202	-1.51
3.760	-0.4316	-0.7231	-1.49
3.765	-0.4289	-0.7244	-1.49
3.770	-0.4253	-0.7246	-1.51
3.775	-0.4222	-0.7234	-1.54
3.780	-0.4201	-0.7200	-1.58
3.785	-0.4193	-0.7160	-1.62
3.790	-0.4201	-0.7120	-1.65
3.795	-0.4224	-0.7088	-1.67

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.800	-0.4255	-0.7068	-1.67
3.805	-0.4282	-0.7049	-1.67
3.810	-0.4313	-0.7038	-1.67
3.815	-0.4341	-0.7031	-1.66
3.820	-0.4367	-0.7028	-1.65
3.825	-0.4394	-0.7025	-1.63
3.830	-0.4418	-0.7031	-1.61
3.835	-0.4423	-0.7020	-1.62
3.840	-0.4440	-0.7058	-1.58
3.845	-0.4453	-0.7058	-1.57
3.850	-0.4466	-0.7051	-1.57
3.855	-0.4478	-0.7046	-1.57
3.860	-0.4493	-0.7040	-1.56
3.865	-0.4510	-0.7032	-1.56
3.870	-0.4521	-0.7026	-1.56
3.875	-0.4539	-0.7020	-1.56
3.880	-0.4555	-0.7013	-1.55
3.885	-0.4571	-0.7004	-1.55
3.890	-0.4590	-0.6999	-1.55
3.895	-0.4610	-0.6989	-1.54
3.900	-0.4624	-0.6980	-1.54
3.905	-0.4645	-0.6968	-1.54
3.910	-0.4661	-0.6961	-1.54
3.915	-0.4683	-0.6952	-1.53
3.920	-0.4704	-0.6943	-1.53
3.925	-0.4723	-0.6934	-1.53
3.930	-0.4740	-0.6923	-1.52
3.935	-0.4758	-0.6910	-1.53
3.940	-0.4778	-0.6901	-1.52
3.945	-0.4797	-0.6890	-1.52
3.950	-0.4816	-0.6879	-1.52
3.955	-0.4834	-0.6864	-1.52
3.960	-0.4852	-0.6850	-1.52
3.965	-0.4870	-0.6840	-1.52
3.970	-0.4887	-0.6823	-1.52
3.975	-0.4906	-0.6809	-1.52
3.980	-0.4925	-0.6796	-1.52
3.985	-0.4941	-0.6784	-1.52
3.990	-0.4960	-0.6765	-1.53
3.995	-0.4979	-0.6750	-1.53
4.000	-0.4997	-0.6736	-1.53
4.005	-0.5014	-0.6721	-1.53
4.010	-0.5030	-0.6705	-1.53
4.015	-0.5046	-0.6689	-1.54
4.020	-0.5064	-0.6673	-1.54
4.025	-0.5079	-0.6661	-1.54

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.030	-0.5097	-0.6641	-1.54
4.035	-0.5113	-0.6625	-1.55
4.040	-0.5129	-0.6609	-1.55
4.045	-0.5143	-0.6593	-1.55
4.050	-0.5156	-0.6574	-1.56
4.055	-0.5173	-0.6558	-1.56
4.060	-0.5185	-0.6543	-1.57
4.065	-0.5198	-0.6530	-1.57
4.070	-0.5211	-0.6514	-1.57
4.075	-0.5223	-0.6499	-1.58
4.080	-0.5238	-0.6485	-1.58
4.085	-0.5248	-0.6473	-1.58
4.090	-0.5261	-0.6459	-1.59
4.095	-0.5274	-0.6446	-1.59
4.100	-0.5283	-0.6432	-1.59
4.105	-0.5296	-0.6422	-1.59
4.110	-0.5307	-0.6408	-1.60
4.115	-0.5322	-0.6398	-1.60
4.120	-0.5335	-0.6389	-1.59
4.125	-0.5349	-0.6378	-1.59
4.130	-0.5364	-0.6366	-1.59
4.135	-0.5377	-0.6352	-1.60
4.140	-0.5387	-0.6336	-1.60
4.145	-0.5395	-0.6320	-1.61
4.150	-0.5410	-0.6303	-1.61
4.155	-0.5415	-0.6289	-1.62
4.160	-0.5420	-0.6276	-1.63
4.165	-0.5422	-0.6260	-1.64
4.170	-0.5424	-0.6248	-1.65
4.175	-0.5420	-0.6241	-1.65
4.180	-0.5421	-0.6237	-1.66
4.185	-0.5420	-0.6227	-1.67
4.190	-0.5419	-0.6223	-1.67
4.195	-0.5418	-0.6219	-1.67
4.200	-0.5415	-0.6216	-1.68
4.205	-0.5417	-0.6215	-1.68
4.210	-0.5415	-0.6211	-1.68
4.215	-0.5409	-0.6213	-1.68
4.220	-0.5408	-0.6213	-1.68
4.225	-0.5408	-0.6214	-1.68
4.230	-0.5404	-0.6212	-1.69
4.235	-0.5404	-0.6212	-1.69
4.240	-0.5401	-0.6216	-1.69
4.245	-0.5397	-0.6221	-1.69
4.250	-0.5395	-0.6223	-1.69
4.255	-0.5391	-0.6230	-1.68

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.260	-0.5389	-0.6231	-1.68
4.265	-0.5384	-0.6236	-1.68
4.270	-0.5380	-0.6238	-1.68
4.275	-0.5378	-0.6241	-1.68
4.280	-0.5374	-0.6245	-1.68
4.285	-0.5372	-0.6253	-1.68
4.290	-0.5367	-0.6254	-1.68
4.295	-0.5367	-0.6260	-1.68
4.300	-0.5362	-0.6263	-1.68
4.305	-0.5357	-0.6269	-1.67
4.310	-0.5356	-0.6275	-1.67
4.315	-0.5354	-0.6279	-1.67
4.320	-0.5352	-0.6285	-1.67
4.325	-0.5349	-0.6292	-1.66
4.330	-0.5347	-0.6298	-1.66
4.335	-0.5345	-0.6306	-1.65
4.340	-0.5343	-0.6312	-1.65
4.345	-0.5342	-0.6321	-1.64
4.350	-0.5338	-0.6330	-1.64
4.355	-0.5335	-0.6339	-1.63
4.360	-0.5331	-0.6347	-1.63
4.365	-0.5326	-0.6357	-1.63
4.370	-0.5321	-0.6365	-1.62
4.375	-0.5317	-0.6375	-1.62
4.380	-0.5310	-0.6383	-1.61
4.385	-0.5304	-0.6390	-1.61
4.390	-0.5296	-0.6400	-1.61
4.395	-0.5291	-0.6408	-1.61
4.400	-0.5286	-0.6412	-1.61
4.405	-0.5281	-0.6418	-1.61
4.410	-0.5275	-0.6423	-1.61
4.415	-0.5266	-0.6426	-1.61
4.420	-0.5266	-0.6430	-1.61
4.425	-0.5257	-0.6434	-1.61
4.430	-0.5253	-0.6435	-1.61
4.435	-0.5253	-0.6435	-1.61
4.440	-0.5251	-0.6436	-1.61
4.445	-0.5248	-0.6438	-1.61
4.450	-0.5248	-0.6443	-1.61
4.455	-0.5248	-0.6448	-1.60
4.460	-0.5246	-0.6452	-1.60
4.465	-0.5242	-0.6463	-1.60
4.470	-0.5237	-0.6474	-1.59
4.475	-0.5234	-0.6484	-1.58
4.480	-0.5227	-0.6500	-1.58
4.485	-0.5230	-0.6513	-1.56

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.490	-0.5225	-0.6530	-1.55
4.495	-0.5226	-0.6542	-1.54
4.500	-0.5220	-0.6554	-1.54
4.505	-0.5215	-0.6562	-1.53
4.510	-0.5210	-0.6570	-1.53
4.515	-0.5205	-0.6569	-1.53
4.520	-0.5199	-0.6573	-1.53
4.525	-0.5195	-0.6573	-1.54
4.530	-0.5190	-0.6564	-1.55
4.535	-0.5188	-0.6559	-1.55
4.540	-0.5189	-0.6554	-1.56
4.545	-0.5191	-0.6545	-1.56
4.550	-0.5193	-0.6541	-1.56
4.555	-0.5196	-0.6536	-1.57
4.560	-0.5201	-0.6530	-1.57
4.565	-0.5204	-0.6525	-1.57
4.570	-0.5205	-0.6515	-1.58
4.575	-0.5210	-0.6507	-1.58
4.580	-0.5213	-0.6502	-1.58
4.585	-0.5212	-0.6495	-1.59
4.590	-0.5220	-0.6483	-1.59
4.595	-0.5226	-0.6474	-1.60
4.600	-0.5229	-0.6467	-1.60
4.605	-0.5236	-0.6455	-1.61
4.610	-0.5241	-0.6448	-1.61
4.615	-0.5246	-0.6441	-1.61
4.620	-0.5251	-0.6431	-1.62
4.625	-0.5255	-0.6424	-1.62
4.630	-0.5257	-0.6412	-1.63
4.635	-0.5259	-0.6403	-1.63
4.640	-0.5262	-0.6392	-1.64
4.645	-0.5266	-0.6385	-1.64
4.650	-0.5269	-0.6376	-1.65
4.655	-0.5268	-0.6365	-1.66
4.660	-0.5271	-0.6363	-1.66
4.665	-0.5270	-0.6350	-1.67
4.670	-0.5271	-0.6344	-1.67
4.675	-0.5269	-0.6334	-1.68
4.680	-0.5266	-0.6329	-1.69
4.685	-0.5263	-0.6321	-1.70
4.690	-0.5263	-0.6316	-1.70
4.695	-0.5259	-0.6309	-1.71
4.700	-0.5258	-0.6304	-1.71
4.705	-0.5258	-0.6293	-1.72
4.710	-0.5257	-0.6287	-1.73
4.715	-0.5257	-0.6279	-1.73

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.720	-0.5260	-0.6269	-1.74
4.725	-0.5263	-0.6260	-1.75
4.730	-0.5266	-0.6249	-1.75
4.735	-0.5272	-0.6239	-1.76
4.740	-0.5281	-0.6223	-1.76
4.745	-0.5292	-0.6211	-1.77
4.750	-0.5297	-0.6199	-1.77
4.755	-0.5308	-0.6184	-1.78
4.760	-0.5319	-0.6171	-1.78
4.765	-0.5324	-0.6164	-1.78
4.770	-0.5328	-0.6155	-1.79
4.775	-0.5328	-0.6145	-1.79
4.780	-0.5326	-0.6135	-1.80
4.785	-0.5326	-0.6133	-1.81
4.790	-0.5316	-0.6125	-1.82
4.795	-0.5308	-0.6116	-1.83
4.800	-0.5301	-0.6110	-1.84
4.805	-0.5292	-0.6105	-1.85
4.810	-0.5284	-0.6098	-1.86
4.815	-0.5271	-0.6091	-1.88
4.820	-0.5265	-0.6087	-1.89
4.825	-0.5258	-0.6086	-1.89
4.830	-0.5252	-0.6081	-1.90
4.835	-0.5250	-0.6082	-1.90
4.840	-0.5245	-0.6082	-1.90
4.845	-0.5247	-0.6084	-1.90
4.850	-0.5249	-0.6083	-1.90
4.855	-0.5250	-0.6091	-1.89
4.860	-0.5250	-0.6091	-1.89
4.865	-0.5247	-0.6098	-1.89
4.870	-0.5255	-0.6101	-1.88
4.875	-0.5260	-0.6106	-1.87
4.880	-0.5267	-0.6111	-1.87
4.885	-0.5264	-0.6121	-1.86
4.890	-0.5275	-0.6124	-1.85
4.895	-0.5279	-0.6127	-1.84
4.900	-0.5285	-0.6131	-1.84
4.905	-0.5288	-0.6136	-1.83
4.910	-0.5295	-0.6138	-1.82
4.915	-0.5301	-0.6143	-1.81
4.920	-0.5307	-0.6145	-1.81
4.925	-0.5310	-0.6147	-1.81
4.930	-0.5317	-0.6152	-1.80
4.935	-0.5323	-0.6155	-1.79
4.940	-0.5329	-0.6156	-1.79
4.945	-0.5334	-0.6158	-1.78

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.950	-0.5338	-0.6162	-1.77
4.955	-0.5343	-0.6164	-1.77
4.960	-0.5349	-0.6164	-1.76
4.965	-0.5352	-0.6166	-1.76
4.970	-0.5358	-0.6170	-1.75
4.975	-0.5362	-0.6173	-1.75
4.980	-0.5369	-0.6175	-1.74
4.985	-0.5373	-0.6175	-1.74
4.990	-0.5379	-0.6180	-1.73
4.995	-0.5383	-0.6182	-1.73
5.000	-0.5390	-0.6182	-1.72
5.005	-0.5394	-0.6185	-1.72
5.010	-0.5400	-0.6184	-1.71
5.015	-0.5403	-0.6186	-1.71
5.020	-0.5408	-0.6182	-1.71
5.025	-0.5409	-0.6182	-1.71
5.030	-0.5413	-0.6177	-1.71
5.035	-0.5409	-0.6180	-1.71
5.040	-0.5412	-0.6169	-1.72
5.045	-0.5408	-0.6174	-1.72
5.050	-0.5403	-0.6167	-1.73
5.055	-0.5400	-0.6176	-1.72
5.060	-0.5396	-0.6167	-1.73
5.065	-0.5395	-0.6175	-1.72
5.070	-0.5394	-0.6174	-1.73
5.075	-0.5402	-0.6175	-1.72
5.080	-0.5399	-0.6176	-1.72
5.085	-0.5408	-0.6172	-1.72
5.090	-0.5413	-0.6175	-1.71
5.095	-0.5427	-0.6167	-1.71
5.100	-0.5430	-0.6171	-1.70
5.105	-0.5443	-0.6169	-1.70
5.110	-0.5453	-0.6164	-1.69
5.115	-0.5463	-0.6156	-1.69
5.120	-0.5473	-0.6155	-1.69
5.125	-0.5483	-0.6143	-1.69
5.130	-0.5494	-0.6140	-1.68
5.135	-0.5501	-0.6130	-1.68
5.140	-0.5514	-0.6122	-1.68
5.145	-0.5524	-0.6112	-1.68
5.150	-0.5539	-0.6101	-1.68
5.155	-0.5549	-0.6087	-1.68
5.160	-0.5562	-0.6074	-1.69
5.165	-0.5573	-0.6060	-1.69
5.170	-0.5587	-0.6045	-1.69
5.175	-0.5604	-0.6024	-1.69

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.180	-0.5613	-0.6010	-1.70
5.185	-0.5628	-0.5990	-1.70
5.190	-0.5638	-0.5970	-1.71
5.195	-0.5648	-0.5950	-1.72
5.200	-0.5657	-0.5926	-1.73
5.205	-0.5668	-0.5910	-1.74
5.210	-0.5681	-0.5889	-1.74
5.215	-0.5690	-0.5865	-1.75
5.220	-0.5702	-0.5842	-1.76
5.225	-0.5710	-0.5819	-1.77
5.230	-0.5722	-0.5796	-1.78
5.235	-0.5734	-0.5775	-1.79
5.240	-0.5744	-0.5752	-1.80
5.245	-0.5754	-0.5732	-1.81
5.250	-0.5764	-0.5707	-1.82
5.255	-0.5773	-0.5686	-1.83
5.260	-0.5783	-0.5663	-1.84
5.265	-0.5796	-0.5645	-1.84
5.270	-0.5806	-0.5623	-1.85
5.275	-0.5815	-0.5602	-1.86
5.280	-0.5825	-0.5582	-1.87
5.285	-0.5834	-0.5562	-1.87
5.290	-0.5845	-0.5544	-1.88
5.295	-0.5851	-0.5529	-1.88
5.300	-0.5861	-0.5509	-1.89
5.305	-0.5871	-0.5492	-1.90
5.310	-0.5879	-0.5473	-1.90
5.315	-0.5887	-0.5459	-1.91
5.320	-0.5891	-0.5438	-1.92
5.325	-0.5900	-0.5426	-1.92
5.330	-0.5906	-0.5408	-1.93
5.335	-0.5913	-0.5389	-1.94
5.340	-0.5918	-0.5375	-1.94
5.345	-0.5926	-0.5361	-1.95
5.350	-0.5929	-0.5344	-1.96
5.355	-0.5936	-0.5325	-1.97
5.360	-0.5941	-0.5304	-1.98
5.365	-0.5946	-0.5290	-1.98
5.370	-0.5954	-0.5271	-1.99
5.375	-0.5963	-0.5253	-2.00
5.380	-0.5969	-0.5237	-2.00
5.385	-0.5981	-0.5224	-2.00
5.390	-0.5990	-0.5209	-2.01
5.395	-0.5998	-0.5193	-2.01
5.400	-0.6006	-0.5181	-2.01
5.405	-0.6016	-0.5170	-2.01

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.410	-0.6028	-0.5158	-2.01
5.415	-0.6033	-0.5143	-2.02
5.420	-0.6043	-0.5133	-2.02
5.425	-0.6053	-0.5123	-2.01
5.430	-0.6059	-0.5109	-2.02
5.435	-0.6068	-0.5099	-2.02
5.440	-0.6078	-0.5092	-2.02
5.445	-0.6082	-0.5083	-2.02
5.450	-0.6093	-0.5077	-2.01
5.455	-0.6098	-0.5067	-2.02
5.460	-0.6103	-0.5061	-2.02
5.465	-0.6109	-0.5052	-2.02
5.470	-0.6117	-0.5043	-2.02
5.475	-0.6121	-0.5037	-2.02
5.480	-0.6130	-0.5029	-2.02
5.485	-0.6136	-0.5019	-2.02
5.490	-0.6147	-0.5012	-2.01
5.495	-0.6153	-0.5005	-2.01
5.500	-0.6163	-0.5001	-2.01
5.505	-0.6173	-0.4996	-2.00
5.510	-0.6181	-0.4993	-2.00
5.515	-0.6190	-0.4989	-1.99
5.520	-0.6200	-0.4988	-1.98
5.525	-0.6208	-0.4993	-1.97
5.530	-0.6217	-0.4991	-1.97
5.535	-0.6225	-0.4991	-1.96
5.540	-0.6232	-0.4992	-1.95
5.545	-0.6239	-0.4992	-1.95
5.550	-0.6245	-0.4994	-1.94
5.555	-0.6251	-0.4995	-1.94
5.560	-0.6258	-0.4995	-1.93
5.565	-0.6267	-0.4997	-1.92
5.570	-0.6274	-0.4998	-1.91
5.575	-0.6281	-0.4998	-1.91
5.580	-0.6290	-0.4999	-1.90
5.585	-0.6297	-0.4999	-1.89
5.590	-0.6307	-0.4999	-1.89
5.595	-0.6316	-0.4998	-1.88
5.600	-0.6327	-0.5000	-1.87
5.605	-0.6336	-0.5002	-1.86
5.610	-0.6347	-0.5003	-1.85
5.615	-0.6356	-0.5005	-1.84
5.620	-0.6365	-0.5009	-1.83
5.625	-0.6375	-0.5013	-1.82
5.630	-0.6384	-0.5015	-1.81
5.635	-0.6390	-0.5017	-1.80

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.640	-0.6397	-0.5017	-1.80
5.645	-0.6403	-0.5021	-1.79
5.650	-0.6411	-0.5023	-1.78
5.655	-0.6414	-0.5024	-1.78
5.660	-0.6422	-0.5026	-1.77
5.665	-0.6426	-0.5027	-1.77
5.670	-0.6432	-0.5025	-1.76
5.675	-0.6436	-0.5027	-1.76
5.680	-0.6446	-0.5025	-1.75
5.685	-0.6445	-0.5024	-1.75
5.690	-0.6449	-0.5019	-1.75
5.695	-0.6451	-0.5019	-1.75
5.700	-0.6457	-0.5017	-1.75
5.705	-0.6457	-0.5014	-1.75
5.710	-0.6461	-0.5013	-1.75
5.715	-0.6459	-0.5011	-1.75
5.720	-0.6464	-0.5007	-1.75
5.725	-0.6466	-0.5006	-1.75
5.730	-0.6473	-0.4999	-1.75
5.735	-0.6473	-0.4998	-1.75
5.740	-0.6477	-0.4998	-1.74
5.745	-0.6480	-0.4996	-1.74
5.750	-0.6475	-0.4988	-1.75
5.755	-0.6477	-0.4982	-1.75
5.760	-0.6480	-0.4974	-1.76
5.765	-0.6476	-0.4966	-1.77
5.770	-0.6476	-0.4958	-1.77
5.775	-0.6479	-0.4955	-1.77
5.780	-0.6480	-0.4942	-1.78
5.785	-0.6483	-0.4933	-1.78
5.790	-0.6480	-0.4922	-1.79
5.795	-0.6480	-0.4911	-1.80
5.800	-0.6479	-0.4897	-1.81
5.805	-0.6476	-0.4884	-1.82
5.810	-0.6473	-0.4873	-1.83
5.815	-0.6472	-0.4862	-1.84
5.820	-0.6474	-0.4849	-1.84
5.825	-0.6477	-0.4833	-1.85
5.830	-0.6474	-0.4821	-1.86
5.835	-0.6470	-0.4808	-1.87
5.840	-0.6469	-0.4794	-1.88
5.845	-0.6474	-0.4779	-1.89
5.850	-0.6471	-0.4767	-1.90
5.855	-0.6471	-0.4752	-1.91
5.860	-0.6471	-0.4736	-1.92
5.865	-0.6477	-0.4723	-1.92

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	-0.6473	-0.4709	-1.93
5.875	-0.6475	-0.4698	-1.94
5.880	-0.6479	-0.4684	-1.94
5.885	-0.6480	-0.4671	-1.95
5.890	-0.6482	-0.4661	-1.96
5.895	-0.6483	-0.4650	-1.96
5.900	-0.6482	-0.4640	-1.97
5.905	-0.6484	-0.4628	-1.97
5.910	-0.6485	-0.4623	-1.98
5.915	-0.6484	-0.4611	-1.99
5.920	-0.6485	-0.4599	-1.99
5.925	-0.6481	-0.4590	-2.00
5.930	-0.6481	-0.4582	-2.01
5.935	-0.6481	-0.4570	-2.01
5.940	-0.6481	-0.4558	-2.02
5.945	-0.6481	-0.4551	-2.03
5.950	-0.6483	-0.4538	-2.03
5.955	-0.6483	-0.4530	-2.04
5.960	-0.6487	-0.4518	-2.04
5.965	-0.6493	-0.4507	-2.04
5.970	-0.6498	-0.4503	-2.04
5.975	-0.6500	-0.4497	-2.04
5.980	-0.6506	-0.4494	-2.04
5.985	-0.6510	-0.4492	-2.04
5.990	-0.6517	-0.4490	-2.03
5.995	-0.6519	-0.4493	-2.03
6.000	-0.6523	-0.4495	-2.02

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to ±17.2 at 5 MHz … ±2.1 at 40 MHz per σ (usable: $\leq$ ±2). Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable ($|S11_ref1 - S11_ref2| < 1e\text{-}02$). Consider starting the sweep above the affected range.